

Figure 8.4. Supply and long-run equilibrium in the pfillip market when the price of kapitose rises with increasing demand for kapitose by pfillip producers.

(Don't be lazy; go through all the steps in these math derivations!) Hence if q is the long-run equilibrium price of kapitose, $(2q)^{1/6}$ is the long-run equilibrium price of pfillip. Turning this around, if p is the long-run price of pfillip, the equilibrium price of kapitose must be $q(p) = p^6/2$.

At the same time, if $q = p^6/2$ is the price of kapitose and if each firm in the pfillip industry is producing optimally at the efficient scale of $y^e = 1/[3(2q)^{1/6}] = 1/(3p)$, each firm is using $k^e = y^2/(2q)^{2/3} = 1/(18q) = 1/(9p^6)$ units of kapitose. (You need to work out the solution to the cost minimization problem above to see how we did this.)

This may seem counterintuitive to you. We have found that the level of production of the single firm is a *decreasing* function of the price of pfillip! But this is not counterintuitive as long as you remember that we are speaking here of equilibrium values. The only way the price of pfillip could be higher in a long-run equilibrium is if the price of kapitose is higher. And a higher price of kapitose lowers the efficient scale of all the firms.

Suppose long-run industry supply of pfillip is S when the price of pfillip is p . Since each firm is producing $1/(3p)$, this means we have $J = 3pS$ firms. Each of these firms uses $1/(9p^6)$ units of kapitose, so total kapitose utilization by the pfillip industry firms is

$$K = \frac{J}{9p^6} = \frac{3pS}{9p^6} = \frac{S}{3p^5}.$$

This causes the price of kapitose to be

$$\frac{p^6}{2} = q = \frac{K}{200} = \frac{S}{200(3p^5)}.$$

Solving for S in terms of p yields

$$S(p) = 300p^{11}.$$

This is the long-run equilibrium supply curve for pfillip.

In figure 8.4 we've drawn what we've discovered. The two demand curves are the two curves from the previous example: the original $D(p) = 400 - 100p$; and the shifted demand curve $D(p) = 750 - 150p$. We've dashed in the long-run supply curve of pfillip under the conditions that the price of kapitose is always \$.50; this is perfectly elastic supply at $p = 1$ since, if the price of kapitose never changes, the long-run average cost curves of firms are always the same (at any scale of industry production), and so minimum average cost is always \$1. And we've drawn in the long-run industry supply curve for the case where the price of kapitose rises with kapitose demand; this is the curve $S(p)$ derived above.

Note that the two supply curves intersect at $p = 1$, at the level $S(p) = 300$, which just happens to be the equilibrium quantity for both with the original demand curve. This is no coincidence; the numbers were picked so this would happen.^c The point is that at higher levels of long-run equilibrium price, the long-run industry supply of pfillip is less than when kapitose costs a flat \$.50. Hence if demand shifts in the fashion of the example and the price of kapitose rises with kapitose usage, the new long-run equilibrium, marked with a * in the figure, has a price higher than \$1 and an equilibrium quantity less than the 600 we had before. (It is left to you to compute the new long-run equilibrium prices and quantities. Since this involves an eleventh degree polynomial, you will probably have to resort to numerical approximation!)

This derivation may seem quite involved to you. It is. But it has to be. We are solving for equilibria in two markets at once. We can't tell what the price of kapitose will be until we know how much is demanded by the pfillip producers. But we can't tell how much is demanded by pfillip producers until we know the price of kapitose. And we are relying on the pfillip market being in a long-run equilibrium. Putting all that together is not an easy exercise. (Now try problem 3.)

One point should be stressed about the example. The individual firm is always acting as a price taker. When we solved the individual firm's cost minimization problem, we took the price of kapitose to be an unchanging q . As we noted at the start of the chapter, this isn't quite right. As even a single firm increases its demand for kapitose, the price of kapitose rises. But this is no worse than what happens in the individual firm's profit-maximization problem, given price p . Given downward sloping demand, as a single firm raises its level of output, it depresses the market price however slightly, something else our price-taking firms are ignoring.

8.4. General equilibrium with firms

Since perfectly competitive markets may be tied together as in the previous section, we might try to put competitive firms into a general equilibrium analysis.

^c It wasn't hard to do. Knowing that the first equilibrium has $p = 1$, $q = .5$, $Y = 300$, and industry kapitose utilization of $K = 100$, a supply curve for kapitose was selected so that at $K = 100$, q would be .5.

This is not an obvious thing to do if you are doing it to study the behavior of the economy we inhabit. A model of general equilibrium in which all firms are price takers is not all that realistic. Some firms, and some industries, may be perfectly competitive (or may be nearly so), but we might be sceptical if it is asserted that all are. That is one of the virtues of partial equilibrium analysis; if one makes an assumption such as firms are competitive, the assumption doesn't have to apply to every firm, but only to those currently under examination.

At the same time, we have the problems of partial equilibrium analysis highlighted in the previous section to contend with. We can do as we did in the example: Try to cobble together a model of all the markets that are relatively strongly connected, without bringing into the picture absolutely every market there is. Or, if we are really interested in studying the general equilibrium, we can "hold out" the production plans of firms that are not competitive. Recall the definition we gave in section 6.2 of an exchange equilibrium with fixed production. We could try a similar trick here: Take as exogenously given the production plans of firms that are not price takers (and, if it exists, any government or quasi-government production), and endogenously model the production decisions of competitive firms, along the lines to be sketched below.

We will not give a lot of details here, but instead give a quick overview of what goes on in general equilibrium with firms.

The economy

An economy is specified by:

- (a) A finite number K of commodities.
- (b) A finite number J of firms. Each firm j is specified by a production possibility set $Z^j \subseteq R^K$.
- (c) A finite number I of consumers. We assume that each consumer can consume any nonnegative bundle of the K goods, so the consumer's preferences are defined on the positive orthant of R^K , denoted X . These preferences are assumed to be representable by a utility function $U_i: X \rightarrow R$. We assume throughout that preferences are continuous and are locally insatiable. Each consumer comes with an endowment $e^i \in X$. Each consumer also comes with an entitlement to share in the profits of the firms. (This is a strictly capitalist economy.) Consumer i is entitled to the share s^{ij} of the profits of firm j . These shares will be assumed to be nonnegative and to satisfy $\sum_{i=1}^I s^{ij} = 1$ for each j ; that is, consumers taken together are entitled to all the profits generated by each firm (and no more).

An ensemble of the items listed above is called a (general equilibrium) economy.

Walrasian equilibrium

Definition. A Walrasian equilibrium for a given economy consists of a price vector $p \in R^K$, an array of production plans (z^j) , one for each firm j , and an array of consumption plans (x^i) , one for each consumer i , such that

- (a) For each firm j , $z^j \in Z^j$ (the production plan of j is feasible), and z^j solves the problem

$$\text{maximize } p \cdot z \text{ subject to } z \in Z^j.$$

- (b) For each consumer i , $x^i \in X$, and x^i solves the problem

$$\text{maximize } U_i(x) \text{ subject to } x \in X, p \cdot x \leq p \cdot e^i + \sum_{j=1}^J s^{ij} p \cdot z^j.$$

- (c) Markets clear: $\sum_{i=1}^I x^i \leq \sum_{i=1}^I e^i + \sum_{j=1}^J z^j$.

Or in words, firms are maximizing profits, consumers are maximizing their utility given their budget constraints, and demand is less than supply for all commodities. Note especially the budget constraint for the consumer; the consumer derives wealth from her endowment and from the profits paid out by firms.^d

Three technical remarks are in order:

- (1) It should be obvious that if $(p, (z^j), (x^i))$ is a Walrasian equilibrium, so is $(\lambda p, (z^j), (x^i))$, for any scalar $\lambda > 0$.
- (2) We will want (for reasons of ease of exposition) to conclude that equilibrium prices are nonnegative. In chapter 6 we obtained this conclusion by assuming that preferences are nondecreasing. That will work quite well here. But we can equally well assume that at least one firm has access to a free disposal technology. (The proof is left to you.) We hereafter will assume that equilibrium prices are always nonnegative, by virtue of one of these two reasons. Of course, local insatiability implies that equilibrium prices could not be identically zero.

^d It has not been assumed that profits are nonnegative; consumers are assumed to make up the losses incurred by a firm in proportion to their shareholdings. But by adding an assumption that $0 \in Z^j$ for each j , this unhappy possibility is avoided.

(3) Since consumers are locally insatiable, we know that in any equilibrium they will spend their entire budget, or $p \cdot x^i = p \cdot e^i + \sum_{j=1}^J s^{ij} p \cdot z^j$. Sum this over i using the fact that $\sum_{i=1}^I s^{ij} = 1$ and you will conclude that

$$p \sum_{i=1}^I x^i = p \cdot e + p \cdot \sum_{j=1}^J z^j,$$

where $e = \sum_{i=1}^I e^i$ is the social endowment. By virtue of (2) then, any good in excess supply in equilibrium must have zero price.

Everything we said in chapter 6 about the credibility of this reduced form solution concept extends to a setting with firms. We haven't specified a trading mechanism or trading institutions, so this is a reduced form solution concept. There are many realistic reasons to doubt that the outcome of economic activity in an economy would approximate a Walrasian equilibrium. But there is experimental evidence to support the concept, at least in very carefully controlled conditions.

The welfare theorems

When considering the efficiency of a Walrasian equilibrium, we are concerned only with the preferences of the consumers. Accordingly, for a fixed general equilibrium economy, the following definitions are made:

Definitions. (a) A *plan* for the economy consists of an array of production plans (z^j) , one for each firm, and an array of consumption plans (x^i) , one for each consumer. A plan $((z^j), (x^i))$ is called *feasible* if $z^j \in Z^j$ for each j , $x^i \in X$ for each i , and $\sum_{i=1}^I x^i \leq \sum_{i=1}^I e^i + \sum_{j=1}^J z^j$.

(b) A feasible plan $((z^j), (x^i))$ is *Pareto efficient* if there is no other feasible plan $((\hat{z}^j), (\hat{x}^i))$ such that the allocation $(\hat{x}^i)$ Pareto dominates (x^i) .

Theorem. The First Theorem of Welfare Economics. If $(p, (z^j), (x^i))$ is a Walrasian equilibrium, then $((z^j), (x^i))$ is a Pareto efficient plan.

The proof is not much different from the proof for a pure exchange economy. Assume that $(p, (z^j), (x^i))$ is a Walrasian equilibrium and that there is some feasible plan $((\hat{z}^j), (\hat{x}^i))$ such that the allocation $(\hat{x}^i)$ Pareto dominates (x^i) . Since the hat plan is feasible, $\sum_{i=1}^I \hat{x}^i \leq \sum_{i=1}^I e^i + \sum_{j=1}^J \hat{z}^j$. Since prices p are nonnegative, we can premultiply this vector inequality with p and keep the inequality:

$$\sum_{i=1}^I p \cdot \hat{x}^i \leq \sum_{i=1}^I p \cdot e^i + \sum_{j=1}^J p \cdot \hat{z}^j.$$

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As noted above, since consumers are locally insatiable,

$$\sum_{i=1}^I p \cdot x^i = \sum_{i=1}^I p \cdot e^i + \sum_{j=1}^J p \cdot z^j.$$

By the argument from chapter 6, we know that $p \cdot x^i \leq p \cdot \hat{x}^i$ for every i , with a strict inequality for at least one consumer i . So

$$\sum_{i=1}^I p \cdot e^i + \sum_{j=1}^J p \cdot z^j = \sum_{i=1}^I p \cdot x^i < \sum_{i=1}^I p \cdot \hat{x}^i \leq \sum_{i=1}^I p \cdot e^i + \sum_{j=1}^J p \cdot \hat{z}^j.$$

Look at the two extreme terms and cancel the common term $\sum_{i=1}^I p \cdot e^i$. This leaves

$$\sum_{j=1}^J p \cdot z^j < \sum_{j=1}^J p \cdot \hat{z}^j.$$

Thus for at least one j , $p \cdot z^j < p \cdot \hat{z}^j$. But this would say that z^j is not a profit-maximizing plan for this firm j at prices p , a contradiction to the definition of a Walrasian equilibrium.

Theorem. The Second Theorem of Welfare Economics. Assume that preferences are convex, continuous, nondecreasing and locally insatiable, and production sets are convex. Let $((z^j), (x^i))$ be a Pareto efficient plan such that x_k^i is strictly positive for all i and k . Then the plan $((z^j), (x^i))$ is the plan part of a Walrasian equilibrium, if we first redistribute endowments and shareholdings among consumers.

We will not go into the details of the proof. The proof used in chapter 6 works quite well, except that you have to modify the definition of $X^\dagger$; let

$$X^\dagger = \{x \in R^K : x = \sum_{i=1}^I x^i \text{ for some feasible plan } ((z^j), (x^i))\}.$$

As we noted in chapter 6, the assumption that the allocation part (x^i) of the plan is strictly positive is much too strong, and serious renditions of this result will weaken this assumption.

At the risk of confusing you, we add a remark about efficiency. We suggested earlier that when some firms in a competitive industry have access to a "better" technology than others do, the better firms will (a) make profits and

(b) operate at larger than efficient scale. (You will prove that [a] implies [b] in problem 2.) The same thing can happen if there is a limit to the number of firms that can enter a perfectly competitive industry; i.e., firms in the industry can be making positive profits and, therefore, be producing at larger than efficient scale. (See problem 4 for an example.) How do these "inefficiencies" square with the result of the first welfare theorem? Is there an implicit assumption in the theorem (which guarantees that a Walrasian equilibrium is efficient) that for any production technology Z possessed by any firm, there will always be a queue of potential entrants possessing this technology? Is it implied that firms in a Walrasian equilibrium always make zero profits?

The answers to the last two questions are resounding noes. Any confusion on this point arises because "efficiency" is being used in two different ways. Efficiency in the welfare theorems pertains to efficiency relative to the productive capacities of the economy, which are determined (in part) by the number of firms possessing each technology. If, in an equilibrium, a firm is making positive profits and producing at larger than efficient scale, then we would be able to improve (in the sense of Pareto) on the equilibrium plan if we could create more firms with this technology. But this isn't how efficiency is defined in the welfare theorems; efficiency in those theorems is relative to the firms that are present in the given economy.

We can read this distinction in two ways. On the one hand, earlier discussion of firms producing at larger than efficient scale shouldn't be taken to mean that the market system (with price-taking firms) leads to correctable inefficiencies; the market system will be doing as well as it can given the economy's technology. On the other hand, even if all firms in an economy are competitive, one can potentially "improve" on the allocation of goods achieved in a Walrasian equilibrium when there are profit-making firms by transferring the technology of the profit-making firms to others. (This last recommendation is not a serious policy recommendation. One has to worry about the supply by innovators of profitable innovations. If the government immediately transfers any profitable innovation from the innovator to many competitors, there wouldn't be much incentive to innovate. Because these issues are well beyond us at this point, they are left for the interested reader to pursue.)

Externalities, taxes, and efficiency

As we noted in chapter 6, externalities can overthrow the conclusions of the two welfare theorems. That remains as true here as in the pure exchange context, if not more so. In chapter 6 we considered consumption externalities; firms add to the potential for consumption externalities, as anyone who has lived next to an oil refinery or chemical plant can testify.⁷

⁷ We should note that insofar as effluents and noxious odors from refineries and chemical plants are modeled as commodities, we will have to redo a lot of the theory we've presented; these are goods for which preferences are not nondecreasing, and they are not freely disposable by firms. If you read on in general equilibrium theory, you will see how to model noxious commodities like these.

But the presence of firms gives us a new category of externalities to worry about, namely production externalities.

The idea in a production externality is that a firm, by its choice of production plan (or, for that matter, a consumer by her choice of consumption bundle) changes the feasible set of production plans for other firms. A chemical plant sited on a river can certainly change the feasible technology of its downriver neighbor, a soft-drink manufacturer. A firm that extracts oil from one plot of land can affect the ease with which a neighboring firm can extract oil, when the oil comes from a common underground pool. A firm that raises sheep by grazing the sheep on the village green affects the prospects for its neighbor who wishes to graze its goats on the same village green.

When production externalities of these sorts exist, the welfare theorems once again go awry. We will point out where the proof of the first welfare theorem would break down, leaving you the (harder) task of showing that, in such cases, a Walrasian equilibrium can most certainly be inefficient. The theorem breaks down right at the end. Recall that we supposed a feasible plan $((z^j), (\hat{x}^i))$ that was Pareto superior to the Walrasian equilibrium plan $((z^j), (x^i))$. We concluded (and would, absent problems of consumption externalities, continue to conclude) that in this case it would have to be that

$$\sum_{j=1}^J p \cdot \hat{z}^j > \sum_{j=1}^J p \cdot z^j,$$

where p are the Walrasian equilibrium prices. From this we concluded that for at least one firm, $p \cdot \hat{z}^j > p \cdot z^j$, which was taken to imply that z^j was not profit maximizing for firm j at prices p . But when there are production externalities, this last step doesn't quite work. When there are production externalities, what firm j can do depends on the production plans selected by other firms; we would write something like $Z^j((z^{j'}), (x^i))$ for the production possibility set of firm j . And then it might be possible that $\hat{z}^j \notin Z^j((z^{j'}), (x^i))$ while $\hat{z}^j \in Z^j((\hat{z}^{j'}), (\hat{x}^i))$. That is, $\hat{z}^j$ would give better profits for firm j , but isn't feasible given that other firms and consumers are following the plan $((z^{j'}), (x^i))$.

Taxes give further grounds for scepticism about the two welfare theorems. This statement may seem rather perverse, since you might have read the second welfare theorem as saying that to change the distribution of utility among consumers, you should reallocate endowments through a system of taxes and subsidies and then let the market go to work to achieve

efficiency. But you need to be very careful in interpreting the second welfare theorem and the sort of "taxes and subsidies" that go along with it.

In the second welfare theorem, people begin with endowments, and a redistribution of endowments is supposed to be accomplished before any economic activity is undertaken. In contrast, redistributive taxes and transfers in the real economy are undertaken by taxes on the product/outcome of economic activity. You trade leisure, part of your endowment, for wages with which to buy consumption goods, and the chief form of tax in the U.S. economy, an income tax, takes away part of those wages. Hence when you decide how much leisure to trade for wages, you make a marginal calculation based on your after-tax income. Your employer, in deciding how much labor to purchase, calculates on the basis of pretax wages. (Actually, the employer looks at the effect of wages on her post-tax earnings, which compounds many tax effects.) The point is that in the welfare theorems, it is crucial to the results that all consumers and firms face the same set of prices — the same set of marginal tradeoffs. Reallocating endowments at the start doesn't cause problems, as this sort of reallocation doesn't affect the (equilibrium) marginal trade-offs that an individual faces; in particular, all face the same margins. But when we tax wages or impose excise or sales taxes on goods, the prices that the buyer and seller face are forced apart, and there is no particular reason to think that efficiency will result.

Where would this cause the proof of the first welfare theorem to break down? Recall that the proof began with the supposition that $(p, (z^j), (x^i))$ is a Walrasian equilibrium, and $((\hat{z}^j), (\hat{x}^i))$ is a Pareto superior, feasible plan. Using the feasibility constraints and the fact that prices are non-negative, we premultiplied each side of the feasibility constraints by the equilibrium prices to get a "value" inequality for the hat plan and (using local insatiability) a value equation for the equilibrium plan. From there, by comparing the values of bundles and production plans consumer-by-consumer and firm-by-firm we proceeded to derive a contradiction.

What are we to do when different actors in the economy face different prices for the same good? We can still make the firm-by-firm and consumer-by-consumer comparisons using the prices that the firm and consumer face. But we can't use different sets of prices in going from the feasibility inequalities to the value equation/inequality. The entire method of proof breaks down.

There are good reasons to have taxes on endogenous economic variables such as income that are simply missed by the general equilibrium model. Take as given the position that it is equitable to redistribute somewhat from the more fortunate in terms of ability or luck to the less, even, if necessary, at the cost of some loss in efficiency. The second welfare

theorem says that, ideally, you don't need to suffer any efficiency loss; figure out in advance who is among the more fortunate and transfer some of their endowment to the less. But this scheme presumes that the more and less fortunate can be identified at the outset. Endowments and shares in the firms are publicly available data; someone fortunate in these things cannot "hide" her fortune in order to avoid the tax man. In reality, we cannot tell who is more fortunate and who is less — who will be lucky, who has more ability, and so on. The clues we will get will be based on observed market data such as income levels. And, therefore, equitable redistribution may well have to come at the cost of some efficiency loss.

We might try to make the efficiency loss as small as possible, by manipulating the form of taxes we impose. This is taken up in the literature of public finance. Your attention is called especially to the literature on optimal taxation; here a social welfare function is given and, subject to informational constraints that more nearly approximate a realistic setting, the analyst uses the methods of chapter 5 to find an optimal tax system, one which balances the gains from a more equitable distribution of income, which comes out of the social welfare function, with the losses in efficiency, captured by the extent to which the constraints bind.

Existence and local uniqueness

Finally, questions arise about the number of Walrasian equilibria for economies with firms. Recall that in chapter 6, we discussed results along the lines of every exchange equilibrium that meets certain conditions has at least one equilibrium, and "most" exchange economies have only finitely many. Similar results can be obtained for economies with competitive firms. The results on "most" economies having finitely many equilibria involve mathematics that are completely beyond the scope of this book, and we do not comment further, but for readers who read through the subsection on the existence of equilibria in exchange economies, we include some brief technical remarks:

The sort of fixed point techniques sketched in chapter 6 are adapted to the context of economies with competitive firms to prove existence. Besides looking at consumer demand at given prices, we also must look at the firms' profit-maximizing choice of netput vectors. These choices will have to be well behaved mathematically, just as consumers' demand choices are required to be well behaved. But we have already seen that this can be done: If Z is convex, then the set of solutions to the firm's profit-maximization problem at a given set of prices is convex (proposition 7.3[b]); and if Z is closed, the correspondence that associates to each price vector the set of solutions is upper semi-continuous (problem 4 in chapter 7). Actually, a lot of the work that goes on in general equilibrium with firms is done in an attempt to

avoid very strong assumptions on Z that guarantee a solution to the firm's profit-maximization problem for any prices. In particular, general equilibrium theorists like to get existence results even when production technologies have constant returns to scale, where part of the existence proof will be that prices where the firms can make infinite profits will be avoided.

8.5. Bibliographic notes

Virtually every text presents the partial equilibrium analysis of perfectly competitive markets a bit differently from every other text, so you may wish to reinforce the material on partial equilibrium by looking at other treatments. I strongly recommend as well looking at some examples of presumably perfectly competitive industries; as noted in the text, the *Oil Tanker Shipping Industry* case is an example with enough things going on to make the comparison with the theory very interesting.

As for general equilibrium with firms, the standard references given at the end of chapter 6, namely Debreu (1959) and Arrow and Hahn (1971), remain the best places to go.

One of the unsatisfactory things about models of perfect competition is that firms' conjectures are usually wrong; firms assume they have no market power when in fact they have a little bit. One might hope for formal results showing that perfect competition is appropriate as an approximate model of economies where firms have (and realize they have) a small amount of market power. There is a literature on this subject which, if you are very ambitious, you may wish to tackle. You probably should complete part III of this book before looking at this literature. But, if ever and whenever you are ready, a good place to start is with the *Journal of Economic Theory* symposium issue of April 1980.

References

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- Debreu, G. 1959. *Theory of Value*. New Haven: Cowles Foundation.
- Porter, M. 1983. *Cases in Competitive Strategy*. New York: Free Press.

8.6. Problems

- 1. Prove proposition 8.1.
- 2. (a) Prove that it is inconsistent with the theory of perfect competition that a firm in a long-run equilibrium would ever produce a (positive)

quantity at which its long-run average costs are declining. What are the implications of this for an industry in which all firms have average costs that are perpetually declining?

(b) Prove that in any long-run equilibrium, firms produce at efficient scale or larger, and if a firm is making positive profits, it must be producing at larger than efficient scale. (You may assume that average and marginal cost curves are all continuous.)

Endless variations on the example given in section 8.2 and 8.3 can be played. Problem 3 is quite difficult; in it you consider the "equilibrium dynamics" in the second example from the text. In problem 4, you see what happens if there is limited entry into a competitive industry; in problem 5 we consider a case in which firms possess a number of different technologies; and in problem 6 you work through "equilibrium dynamics" when a factor price changes.

■ 3. (a) In the analysis of the example in section 8.3, we worked through the long-run equilibrium supply curve of pfillip for the case where the price of kapitose depends on kapitose demand by pfillip consumers: The price q of kapitose is $K/200$, where K is pfillip industry demand for kapitose as a factor of production. Suppose demand for pfillip is given by $D(p) = 400 - 100p$. What is the long-run equilibrium? (This should be very easy.)

(b) Suppose demand for pfillip is given by $D(p) = 750 - 150p$. What is the long-run equilibrium? (You will probably need to resort to numerical calculations, both here and throughout the rest of the problem.)

(c) Suppose demand for kapitose begins at $D(p) = 400 - 100p$, and the long-run equilibrium you computed in (a) is reached. Then, suddenly, demand for pfillip changes to $D(p) = 750 - 150p$. What is the new short-run equilibrium, where in the short run firms cannot enter or leave the industry and cannot change their levels of kapitose utilization. (Once again, this is very easy, given what is in the text.)

(d) Continuing from (c), what is the equilibrium in the intermediate run, where in the intermediate run firms cannot enter or leave the industry but can change their levels of kapitose and legume utilization. (Good luck!)

■ 4. Imagine that for some reason there can be only 300 firms producing pfillip; only these 300 firms have licenses, say. In the long run, these 300 firms can exit the market (and later reenter, if conditions warrant), and they can vary the amount of legume and kapitose they use. In the intermediate run, firms cannot leave (or enter), but they vary their factor usage. In

the short run, firms can only change their levels of legume usage. Assume the price of kapitose is $q = \$50$, no matter how much kapitose is used by pfillip producers. (That is, we are back in the setting of section 8.2.)

(a) Suppose demand is given by $D(p) = 400 - 100p$. What is the long-run equilibrium in this industry?

(b) Suppose the equilibrium computed in part (a) is reached, and then demand suddenly shifts to $D(p) = 750 - 150p$. What is the short-run equilibrium position? What is the intermediate-run equilibrium position? What is the new, long-run equilibrium position?

(c) What is the long-run industry supply curve?

■ 5. Suppose there are two production processes for making pfillip. In the first, the production function is $y = k^{1/6}l^{1/3}$ and fixed costs are $1/6$. In the second, the production function is $y = k^{1/9}l^{2/9}$ and fixed costs are $1/4$. One hundred firms have access to the first technology, but only those hundred can use this technology. An unlimited number of firms have access to the second technology. Assume that the price of kapitose is $\$50$ no matter how much kapitose is utilized by pfillip producers.

(a) What is the long-run industry supply curve for this industry?

(b) Suppose demand is given by $D(p) = 400 - 100p$. What is the long-run equilibrium for this industry?

(c) Suppose demand suddenly shifts to $D(p) = 750 - 150p$. Work out the short-run, intermediate-run, and new long-run equilibria.

■ 6. In the production of psillip, a nonnarcotic relaxant, there are three factors of production: legume, kapitose, and jehosa fat. Letting l be the amount of legume, k the amount of kapitose, and j the amount of jehosa fat, the production function is

$$y = f(j, k, l) = j^{1/6}k^{1/6}l^{1/6}.$$

The price of legume is a fixed $\$1$ per unit; kapitose costs $\$4$ per unit, and, at the start of this problem, jehosa fat costs $\$2$ per unit. Many firms could enter this industry, all of whom have access to this technology, and each of whom, if active, must pay a fixed cost (in addition to the costs of any factor inputs) of $\$216$. Demand for psillip is given by $D(p) = 1440 - 10p$, where p is the price of psillip.

(a) Assuming conditions of free entry and exit and assuming all firms are price takers in all markets, what is the long-run equilibrium in the psillip

8.6. Problems

market under these conditions? Give: equilibrium price, quantity, number of firms, output per firm, amount of factors used by each firm, profit per firm. (Hint: The equilibrium price is 72 .)

(b) Suddenly the jehosa crop fails, and the price of jehosa fat jumps to $\$4$ per unit. In the very short run, firms can vary the amount of legume they input to production, but the levels of jehosa fat and kapitose are fixed and the number of firms is fixed. In the medium short run, firms can vary the amounts of legume and jehosa fat they use, but the level of kapitose is fixed and the number of firms is fixed. In the medium run, kapitose joins legume and jehosa fat as a variable factor of production, but the number of firms is fixed. And in the long run, there is free entry and exit from this industry. Solve for the very short-run, medium short-run, medium-run and long-run equilibria in this industry, beginning from the equilibrium position of part (a). When you solve for the long-run equilibrium, do not worry about fractional numbers of firms.

The next problem shifts a factor price in the setting of two linked markets.

■ 7. The manufacture of rhillip, a nonnarcotic nutrient, takes two stages. In the first stage, legume and kapitose are used to produce rhipigume. In the second stage, rhipigume and further amounts of kapitose are used to make rhillip. Rhipigume has no uses other than for the manufacture of rhillip; as you know full well by now, legume and kapitose are the bases for many speciality chemicals.

For purposes of this problem, suppose that legume and kapitose are traded in perfectly competitive markets. The price of kapitose is $\$1$, and the price of legume is $\$4$ (for the time being).

Rhipigume is produced in a perfectly competitive industry that is filled with many identical firms, each of which has production function

$$r = k^{1/4}l^{1/4},$$

where r is the amount of rhipigume produced from k units of kapitose and l units of legume. Each firm in this industry faces fixed costs of $\$16$ (per firm) as well. There is completely free entry and exit to and from this industry.

Rhillip is produced in another perfectly competitive industry. (For legal reasons, no firm can produce both rhillip and rhipigume.) There are many firms in this industry as well, each of whom has production function

$$R = k^{1/4}r^{1/4},$$

where R is the amount of rhillip produced from k units of kapitose and r units of rhipigume. Each firm in this industry has fixed costs of \$72 (per firm). There is completely free entry and exit to and from this industry. Industry demand for Rhillip is given by the equation

$$P = 96 - R/10,$$

where P is the price of rhillip and R is the total quantity of rhillip brought to market.

(a) What is the long-run equilibrium in these two industries? I want to know prices for the two goods, numbers of firms in each industry, output supply and factor demands of each firm, profits per firm, and total industry supply and factor demands.

In the short run, firms in both industries can vary the amounts of kapitose they use in production, but not the amounts of legume or rhipigume. In the intermediate run, firms can vary the amounts of their factor inputs, but entry to or exit from either industry is not possible. In the long run, entry and exit becomes possible.

(b) Beginning from the position you computed in part (a), the price of legume suddenly falls to \$.25. What will happen in these two industries in the short run? In the intermediate run? And in the long run? (For the long run, don't worry about fractional numbers of firms.)

The next problem works through what happens when there are constant returns to scale.

■ 8. Suppose that the production function of a single firm in an industry that produces zipple, a nonnarcotic pain killer, out of kapitose and legume is $f(k, l) = k^{1/3}l^{2/3}$. In this case, firms have no fixed costs. Suppose that the price of kapitose is given by $q = 1/2$ and the price of legume is \$1. Suppose there are precisely 100 firms in the industry; they can leave if they are unprofitable, but no one else can enter.

(a) If demand for zipple is given by $D(p) = 400 - 100p$, what is the long-run equilibrium? Are there parts of the long-run equilibrium that are indeterminate? (Yes!)

(b) Demand shifts to $D(p) = 750 - 150p$. What is the new short-run equilibrium where the firms can vary their usage of legumes but not their

usage of kapitose? Does the indeterminacy you found in part (a) affect your answer to this part?

Next we have an easy finger exercise, which will come back to haunt you in Chapter 19.

■ 9. Imagine that there is an industry for a particular product where the demand for this product is given by $P = 100 - X$. There are two factors of production, capital K and labor L ; the price of each will be \$1 throughout. The firms all have identical technology, which is given by a *fixed coefficients* production function:

$$Y = \left(\min \left(\frac{K}{\alpha}, \frac{L}{1-\alpha} \right) \right)^{1/2},$$

where Y is the level of output and α is some constant from $(0, 1)$. Each firm has fixed costs of 16.

(a) Suppose there are precisely six firms in this industry; they can exit if they are unprofitable, but no firms can enter. What is the long-run equilibrium in this case? (Even though there are only six firms, they all entertain competitive conjectures about their effect on various prices.)

(b) Suppose there is free entry into this industry. What is the long-run equilibrium?

■ 10. Suppose one firm in a general equilibrium economy has a technology with free disposal. Prove that Walrasian equilibrium prices are nonnegative.

■ 11. Consider the following economy: There are three goods, legume, tillip and quillip, two consumers (called 1 and 2), and two firms (called x and y). Firm x is owned entirely by consumer 1 and makes tillip out of legume according to the simple linear production technology $t \leq 3l$. That is, for every unit of legume input, this firm produces three times as many (or less) units of tillip. Firm y is owned entirely by consumer 2 and makes quillip out of legume according to the production technology $q = 4l$. Each consumer initially owns 5 units of legume. Consumer 1 has utility function $u_1(t, q) = 6 + .4 \ln(t) + .6 \ln(q)$. Consumer 2 has utility function $u_2(t, q) = 8 + \ln(t) + \ln(q)$.

(a) What is the general equilibrium of this economy? Assume that firms take prices as given and are profit maximizers, and consumers take prices

as given. When you give prices, normalize them so the price of legume is \$1. What would be the general equilibrium if the shareholdings were reversed? If each consumer held a half-share in each firm?

(b) What is the set of all feasible, Pareto efficient allocations for this economy?

chapter nine

Monopoly

The theory of monopoly is, on the face of things, very simple and straightforward. But behind this simple and straightforward theory lie some deep and interesting questions. We won't be able to answer most of these questions here, but we can pose the questions and give a rough idea what the answers might look like. And later in the book we will see some of the answers that have been developed.

9.1. The standard theory

In a monopoly market, we imagine many buyers and a single vendor of a good. This single vendor is called the *monopoly*.¹ Buyers are assumed to be price takers, and their demand as a function of price is given, as in the case of perfect competition, by an aggregate demand function. We will write $D(p)$ for the level of demand at price p , and we will also write $P(d)$ for the *inverse demand function*; that is, $P(D(p)) = p$. To guarantee a well-defined inverse demand function exists, we must make certain assumptions about demand. But we make those without qualms: We assume that demand is downward sloping. In fact, to keep the mathematics simple, we assume that D is a continuously differentiable function whose derivative is strictly negative (and finite) at any argument $p > 0$ such that $D(p) > 0$. To cover "corner cases," we assume that the monopoly can sell as much of the good in question as desired, if the price of the good is set at 0. We do not require that the demand function is continuous at the price of zero; that is, there may be a finite upper bound on the amount demanded at any positive price. And we do not require strictly positive demand at every price; demand may be zero at some finite price (and at all higher prices).

The monopoly is assumed to be characterized by a total cost function $TC(x)$, where x is the level of the monopoly's output. Then, the standard

¹ This is a firm, so we will use the impersonal pronoun *it*. Although it is sometimes awkward, we will refer to the monopoly as if it were making decisions; we should be referring to the monopoly's manager, but that becomes too tortured.

theory runs, the monopoly sets the quantity of output x to maximize its profits:

$$\max_x P(x)x - TC(x),$$

which gives first-order conditions

$$P'(x)x + P(x) = MC(x).$$

That is, we have the slogan that *marginal revenue equals marginal cost*.^a Note here that the monopoly's conjectures about how the price of its product will change with changes in the level of output (or, alternatively, how the level of demand will change with changes in the price he charges) are correct; they are given by the actual market demand curve. Compare this with the perfect competitors of chapter 8, whose conjectures were slightly wrong.

Recall from chapter 7 that we can rewrite marginal revenue as $P(x)(1 + 1/\epsilon(x))$, where $\epsilon(x) = P(x)/(xP'(x))$ is the elasticity of demand. Since $P(x)$ is decreasing (by assumption), ϵ is a negative function. When $-\epsilon(x) < 1$, demand is said to be *inelastic*; when $-\epsilon(x) > 1$, demand is said to be *elastic*; and when $-\epsilon(x) = 1$, demand is said to have *unit elasticity*. From the formula given above for marginal revenue, we see that this means that the monopoly's marginal revenue is negative when demand is inelastic; it is zero when demand has unit elasticity; and it is positive when demand is elastic. Note that whenever marginal revenue is negative, the monopoly increases *revenues* by decreasing quantity sold. Since, it is normally presumed, marginal costs are always positive (or, at least, nonnegative), this implies that a monopoly would never produce a quantity at which demand is inelastic. If marginal costs are everywhere strictly positive, then production would never be at a level where demand has unit elasticity.

We can also think of the monopoly choosing an optimal price p . In this case, the monopoly's problem would be written

$$\max_p D(p)p - TC(D(p)).$$

^a We are sloppy and assume a solution to the monopoly's problem at some finite level of output x . The easiest way to justify this assumption is to assume price is zero beyond some finite level of supply and, as supply goes to zero, prices stay bounded. But we do not restrict attention to this sort of case in examples, and the careful reader will no doubt check us as we proceed. Moreover, we don't check second-order conditions in the marginal revenue equals marginal cost equation. But in examples, TC will be convex and TR will be concave, so we'll always be okay.

9.1. The standard theory

The first-order conditions are

$$D'(p)p + D(p) = MC(D(p))D'(p).$$

If you remember the formula for the derivative of the inverse of a function $f(x)$ is $df^{-1}(y)/dy|_{y^0} = 1/(df/dx)|_{x^0=f^{-1}(y^0)}$, you will see that this is the same first-order equation as before.

There is no presumption that the monopoly will produce at an efficient scale (that is, at a scale where average costs are minimum). In figure 9.1(a), we see a case where the monopoly will produce at an inefficiently large scale, and in figure 9.1(b) a case where the monopoly will produce at an inefficiently small scale.

How do we know this? The problem is to find the position of the marginal revenue curve. You are given a geometrical procedure for doing this in figure 9.2; if you use this procedure, you remember the relative positions of marginal and average cost, and you remember that the monopoly equates marginal revenue to marginal cost, you should have no problem with these conclusions. But this leaves the question: Why is this geometrical procedure correct? Problem 1 asks for a proof.

That's it. That's all there is to the standard theory of monopoly. We think of the monopoly being able to select the quantity it will supply or, equivalently, the price it will charge, and the resulting equilibrium price or quantity is read off the demand curve. But this simple classic theory is subject to a lot of questions, caveats, variations, and elaborations. In the next few sections, a few of these are sketched.

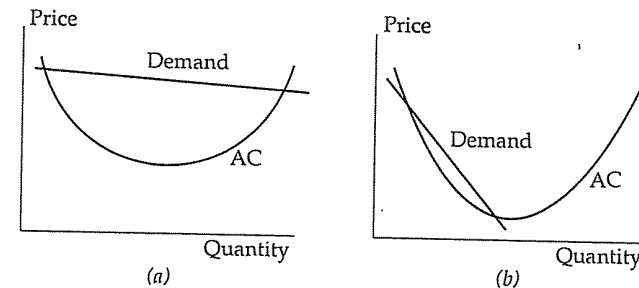


Figure 9.1. The optimal scale of a monopoly.

The optimal scale of a monopoly is inefficiently large in (a) and inefficiently small in (b).

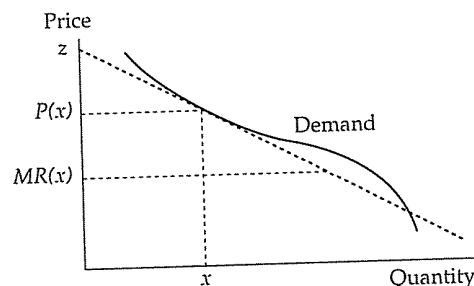


Figure 9.2. Computing the value of marginal revenue.

At any point x along the quantity axis, draw the tangent to the inverse demand demand curve $P(x)$ back to the p -axis. Measure the distance from the point z where this tangent intersects the p -axis to $P(x)$ and move an equal amount on the other side from $P(x)$. This is $MR(x)$. That is, $MR(x) = P(x) - (z - P(x))$.

9.2. Maintaining monopoly

You might wonder, for example, how the monopoly came to be a monopoly, and why it stays that way. If the monopoly is making a profit, why doesn't the industry attract entrants? It may be that legal barriers impede entry. The monopoly might hold a patent on a particular production process or on a particular product. Or it might be illegal for any competitors to enter the market. This is especially likely to happen where the production technology favors one very large company — where average costs are falling at all levels of output. (The case where average costs are decreasing at every level of output is sometimes referred to as the case of a "natural monopoly.") In cases such as this, the monopoly is apt to be subject to regulation; it can't charge any price it wishes, but instead must obtain approval of some regulatory body for the prices it sets. There is a very large literature on the economics of regulation that looks at this situation.

Another reason we might find a monopoly industry is because, while other companies *can* enter this industry, the monopoly acts in a way that forestalls potential competitors. A classic story of this sort associated with the names of Bain and Sylos-Labini runs as follows: Imagine an industry in which there is a constant variable cost of production, say \$1 per unit and a fixed cost for any firm in the industry amounting to \$2.25. (These are convenient, not realistic, numbers.) This production technology is

9.2. Maintaining monopoly

available to one and all. Demand in this industry is given by $D(p) = 9 - p$. There is, at the time being, a single firm in this industry (seemingly a monopoly). Note that the standard theory of monopoly suggests that this monopoly will equate marginal cost, which is 1, with marginal revenue, which is $9 - 2x$ at output level x . (Total Revenue = $9x - x^2$. Differentiate.) This gives $x = (9 - 1)/2 = 4$, $P = 5$, and monopoly profits of $4(5 - 1) - 2.25 = 13.75$.

But despite this prediction of the standard theory, it is discovered upon inspection that the monopoly has set the quantity of output equal to 5, thus realizing a profit of $5(4 - 1) - 2.25 = 12.75$. When asked why, the monopoly responds:

Out there are many people who would enter this industry and take away my profits if they thought that they themselves could make a profit. So put yourself in their shoes. Here am I (the monopoly) with an output of 5. If you enter and have an output of x' , assuming that I keep my output level at 5, the price would be $9 - 5 - x' = 4 - x'$, so your margin (gross of your fixed costs) would be $3 - x'$. You will then (optimally) set $x' = 1.5$ so your gross profits will be 2.25. That is, at best, you will only just cover your fixed costs. Since you won't make a positive profit, you won't enter, and I'll stay a monopoly. On the other hand, if I were producing 4 units, and potential entrants believed that I would hold to this level of output, then they would be able to cover their fixed costs, and they would enter. My choice of an output level of 5 may look silly in the short run, but if it keeps you (and other potential entrants) out of my market, it is a good strategy to employ.

The general idea is that the monopoly's choice of quantity and/or price may be based on a strategy for keeping the monopoly, rather than for maximizing short-term profits.

Now rather a lot is wrong with the story just told, primarily concerning the supposed follow-on if you, as a potential entrant, choose to enter: Why would you suppose that after you enter the monopoly will maintain its quantity of output at 5 units? Why wouldn't it accommodate to your entry? What will it do if you enter? We aren't in a position just yet to talk about such things (since, for one thing, they involve what happens in a duopoly). And so we will leave this simple story of *entry deterrence* as it stands for now, returning to it in chapter 13. But the point remains: If

monopoly power isn't granted by law, then monopolies may take actions that are directed towards the maintenance of monopoly power, which may be very different from short-run profit maximization. Hence, in trying to analyze what a monopoly will do in its market, it is essential to have answers to the questions: Where does the monopoly come from? What (if anything) must be done to maintain it?

Although this may be beating the subject to death, it should also be noted that even when the monopoly results from legal protection, say by a patent, considerations like those just mentioned may intrude. Suppose that a manufacturer does hold an unbreakable patent on a particular product. The scope of patent protection does not always extend to protection against close substitutes for the patented product. If substitute products are produced and sold, they restrain the monopoly's market power by flattening and shifting-in the monopoly's demand curve. That is, as the monopoly raises the price charged, many of its customers will move to the substitute good, and it will see a dramatic fall in the quantity it can sell. In this situation, what is observed will be in accord with the standard theory; it is just that the monopoly will face fairly elastic demand.

But (and this is the reason for introducing this now) suppose those substitute goods are not being produced and sold. Then the monopoly may act in a way to impede the entry of producers of those substitute goods. The story is one degree more complex than the story of entry deterrence sketched above, but the same basic notions apply. What we anticipate seeing in the short run, according to the standard theory, may not be observed, insofar as the monopoly is setting its price to keep substitute products from being placed on the market.

9.3. Multigood monopoly

The idea of substitutes for a monopoly product comes up in another context — that of multigood monopolies. Suppose a monopoly has monopoly rights on two different products that are substitutes. By raising the price of one, demand is stimulated for the other and vice versa. Hence we might well see prices for two substitute goods that are higher than the prices we would expect if we analyzed in isolation the market for either.

An extreme form of this, which combines aspects of the sort of entry deterrence discussed above, involves what are called "sleeping patents." Suppose a firm holds a monopoly in a certain product, and it knows on its own or learns of the existence of a close substitute that is not yet on the market but which may be produced. The monopoly may then seek

to patent (or buy exclusive rights to manufacture and sell) the substitute. And then, once it holds exclusive right to manufacture the substitute, it may produce none at all. Especially when there are large fixed costs in production, the monopoly may maximize its profits by producing only one of several close substitutes. Hence it gains the patent to prevent anyone else from manufacturing the substitute, and then it puts the patent "to sleep" to maximize its overall profits.

A third issue on substitute goods and monopolies concerns the range of product variety. A monopoly often offers a number of varieties of the basic product it produces. In the offering of many varieties, the monopoly is competing with itself; red and blue widgets are probably close substitutes, and insofar as the monopoly is selling red widgets, by offering them in blue it cuts into its own market. There is no real problem if no extra fixed cost is associated with offering multiple colors (or, in general, different varieties). But where there are increased fixed costs, either in manufacture or distribution, then the monopoly must decide how many (and which) varieties to offer. We leave these vague ideas here, although problems 9, 10, and 11 will give you some ideas about how to model and analyze this issue.

On the other hand, a monopoly might hold monopolies in two complementary goods. Lowering the price of one stimulates demand for the other. Hence we might expect to see prices lower than those we would predict from analyzing in isolation the market for either. For example, imagine a firm that produces both cameras and film for a particular sort of photographic process for which it has no rivals. (Think of a case where conventional photography is a substitute but not a perfect substitute.) It isn't hard to imagine that this firm might sell cameras suited for its film at a relatively low price to stimulate demand for its film.

Even more interesting are cases where a monopoly in one product can also participate in a second market in which it doesn't hold a monopoly. Suppose, for example, that there are potential entrants ready and willing to serve the second market. And suppose that the products involved are very strong complements. Then the monopoly may be able to use its monopoly in the first product to gain power in the second; we have what is called an *extending monopoly*. For example, a firm that holds a monopoly on a particular sort of computer might bundle sale of the computer with servicing (which, for the sake of argument, we assume anyone could perform). The company might even go to the extreme of refusing to sell its computers at all — it will only lease them — so that it can control forever who services the machines.

9.4. Nonlinear pricing

The standard theory presumes that the monopoly must charge linear prices. That is, the monopoly sets a price per unit, and customers can buy as many/few units as they wish at that price. The usual justification for this assumption is that the good is bought anonymously and can be resold. If the monopoly were, say, to give a quantity discount, then someone would buy a large amount (at a discount) and resell it in smaller lots to consumers who don't want quite as many. If the monopoly were to try to mark up the price for larger lots, then the consumer desiring a large lot would buy a number of small lots (or have her friends buy small lots and resell them to her). But for many sorts of commodities this justification for linear prices doesn't hold water. The resale of electricity, or telephone services, or education (or just about any sort of service) is difficult if not impossible, and the purchase of electricity or telephone services (at least) is not anonymous. When resale is impossible and/or the purchase of the item is not done anonymously, the monopoly may wish to consider using nonlinear prices to increase its profits.

This is not only true for monopolies. The same sort of consideration might intrude in the case of oligopolists, or even in a competitive industry. But analyses of nonlinear pricing have, for the most part, been devoted to the context of monopoly, because nonlinear pricing is hard enough without introducing considerations of competition.

The absolutely best position for a monopoly to be in is (a) to know the precise utility function of every consumer, (b) to be able to tailor a "price schedule" for each individual consumer, and (c) to be able to control absolutely any resale of the good being sold. Then the monopoly can make a "take-or-leave" offer to each individual consumer, which extracts from the consumer all the surplus that this consumer would otherwise obtain from consumption of the good in question. That is, suppose that our consumer's preferences can be represented by a utility function $u(x, y)$, where x is the amount of the good produced by this monopoly that the consumer consumes, and y is the income that this consumer has to make other purchases. All other prices are being held fixed. Note that this function u is something like an indirect utility function, and it should be parameterized by those fixed other prices. We suppose that this consumer has total income (or wealth) y^0 , so if the consumer consumes none of the monopoly's product, the consumer will have utility $u^0 = u(0, y^0)$. Now imagine that the monopoly makes a take-or-leave offer to the consumer; the consumer may take x^* units of the good, for a payment of z^* in total. Note well: The consumer isn't being given the option of taking as much

9.4. Nonlinear pricing

or as little as he desires at a price per unit of z^*/x^* . The consumer may have x^* or none, and that is it. The consumer, then, will take this offer if it gives him utility greater than u^0 ; that is, if $u(x^*, y^0 - z^*) > u^0$. We can see that the monopoly, if it makes a take-or-leave offer with x^* the amount of the good offered, will set the payment z^* as high as possible while keeping to the constraint $u(x^*, y^0 - z^*) > u^0$. (In mathematical analysis of the problem, we'll have trouble with this strict inequality. So it is typical in economics to rewrite the constraint as $u(x^*, y^0 - z^*) \geq u^0$, and either assume that the consumer, if indifferent, will buy, or later fudge things by assuming that the monopoly charges, say, a nickel less than z^* .)

This is for a single consumer. If we look over all consumers, indexing them by the subscript i , we find that this lucky monopoly is seeking to solve the problem

$$\max \sum_i z_i^* - TC \left(\sum_i x_i^* \right) \quad \text{subject to } u_i(x_i^*, y_i^0 - z_i^*) \geq u_i^0 \text{ for all } i,$$

where TC is the monopoly's total cost function. Letting λ_i be the multiplier on the utility constraint of consumer i , the first-order conditions on z_i^* and x_i^* are

$$1 = \lambda_i \frac{\partial u_i}{\partial y} \bigg|_{(x_i^*, z_i^*)} \quad \text{and} \quad MC \left(\sum_i x_i^* \right) = \lambda_i \frac{\partial u_i}{\partial x} \bigg|_{(x_i^*, z_i^*)}.$$

Combining these two, we see that our monopoly will set each consumer's marginal rate of substitution of the good for money to be equal to its own "marginal rate of substitution," which is just its marginal cost. In other words, the monopoly will make available to every consumer enough of the good so the consumer's marginal value (in terms of dollars of other consumption) for the good is just equal to the monopoly's marginal cost. (How does this compare with the strategy of a monopoly that is using linear prices?) Then the monopoly extracts all the utility gains that the consumer gets by setting the fixed take-or-leave price at the most the consumer is willing to pay. It takes a little more algebra, but it can be demonstrated that, short of coercion, there is no way the monopoly can do better than this.

Of course, the assumptions that underpin this analysis are extreme. The monopoly is assumed to be able to deal with each consumer individually. It knows the utility function of each. It can tailor its prices to each; one consumer might be asked to pay \$100 for 100 units of the good,

whereas a second, who gets greater utility from the consumption of 50 units, might be asked to pay \$500 for those 50. (For these two to be part of an optimal scheme, the marginal valuation of the first consumer for the 101st unit must equal the marginal valuation of the second for the 51st, and each of these must equal the monopoly's marginal cost for producing another unit.)

In fact, it seems unlikely that a monopoly would have this much knowledge and power. Even if the monopoly knows the utility function of each individual consumer, it may be compelled by law to offer to any consumer any deal that it offers to a second. (In which case our second consumer above would, it seems likely, prefer to pay \$100 for 100 units, and forego paying \$500 for 50.) Of course, if the monopoly doesn't know which consumer has which utility function, then it is forced to offer to each one the same set of deals and leave it to them to select which they wish to accept.

Still, even if the same set of deals must be offered to all consumers (allowing them to pick which they want), nonlinear prices may come in handy. We might imagine the monopoly trying linear prices but with a fixed charge to each consumer; you can buy as much as you wish (as a consumer) at a set price p per unit, but if you buy any, you must pay k up front. Or we might imagine a scheme where consumers can buy up to n_1 units at a price of p_1 per unit, and further units cost p_2 apiece. Will p_1 be greater or less than p_2 ? It depends, in part, on how well the monopoly can control to whom it sells how much. You are probably familiar, on the one hand, with quantity discounts for telephone service. On the other hand, the rate of interest on a loan will typically increase with the size of the loan.

Actually solving for the optimal nonlinear pricing scheme can be quite an involved problem. But the basic method is easy to state. One supposes that the population of consumers has a given distribution of utility functions, levels of wealth, and so on. Then, for each pricing scheme, one can work out how each type of consumer will react, hence how consumer reactions will be distributed across the population. From this, one obtains profits for the given scheme. If the scheme is characterized by only a few parameters (e.g., a price plus fixed fee system has two parameters, namely k and p ; a two-part linear tariff system has three, namely p_1 , n_1 and p_2), one might be able to solve analytically for the optimal scheme. Or one could at least conduct a numerical search over values of the parameters. For the optimal nonlinear price scheme in general, somewhat more advanced techniques (the calculus of variations or control theory) may be needed, but the same basic method is employed.

An example: A monopoly selling to monopolies

To illustrate some of these notions of nonlinear pricing and price discrimination, consider the following example. A manufacturing monopoly can produce a particular item, called a poiuyt, at a constant unit cost of c per unit. For some reason this manufacturer cannot engage in the retailing of this item. Instead, it must work through a system of retailers, each of whom holds a monopoly on the retail business in a particular geographical region. More specifically, in each of I regions, indexed by $i = 1, \dots, I$, one firm is allowed to engage in the retail sale of poiuyts. We will assume that retailing is a simple business — so simple that it involves no cost at all, except for the cost incurred by the retailer in buying the units from the manufacturer. Demand for poiuyts at the retail level in region i is given by the demand curve $x_i = (A - p_i)/B_i$ or, equivalently, $p_i = A - B_i x_i$, where p_i is the retail price in region i , and x_i is the level of units sold in region i ; A and B_i are positive constants.² We will assume that the manufacturer sets price(s) for the retailers, who then set prices p_i for the retail trade in each region. To keep matters simple, we'll assume that there is no possibility of customers in one region buying from the retailer or some middleman in another, so there is no need to insist that $p_i = p_j$ in equilibrium. Moreover, we'll assume that, in the retail markets, only linear prices can be charged and no discrimination is possible. We will, however, not *insist* on these requirements for the wholesale markets. Instead, we'll make a series of different assumptions about the wholesale market to see how things change with those assumptions.

To begin, we'll assume that the manufacturer, whom we will call M hereafter (using pronouns she and her, to distinguish from the retailers for whom we use he and him), must use linear prices, but is allowed to discriminate among the various retailers. That is, each retailer i is quoted a wholesale price P_i , which this retailer must pay per unit bought at wholesale. The retailer is allowed to choose how many units he wishes to buy and resell (or, equivalently, what retail price p_i he will charge his customers). Retailer i , facing wholesale price P_i , will treat P_i as his marginal cost and will set p_i to

$$\max_{p_i} \frac{(p_i - P_i)(A - p_i)}{B_i}.$$

² We assume throughout that $A > c$. The fact that the same constant A applies to all regions is an enormous convenience analytically. If you wish to, try replicating the analysis to follow for the case where A_i changes from region to region. If you do, be prepared to do a lot of algebra.

Since this is nearly the first time we've worked with linear demand and imperfect competition, and it is far from the last, let us solve this in gruesome detail. First, since demand is given by $p_i = A - B_i x_i$, total revenue at sales level x_i is $TR(x_i) = Ax_i - B_i x_i^2$. Marginal revenue is thus $A - 2B_i x_i$. Marginal revenue equals marginal costs (of P_i) at $A - 2B_i x_i = P_i$, or $x_i = (A - P_i)/2B_i$, which gives $p_i = (A + P_i)/2$, and profits $\pi_i = (A - P_i)^2/4B_i$. We depict all this in figure 9.3. Demand is given by the solid line. Marginal revenue is the dashed line; note that it has the same intercept on the price axis, and its slope is twice as great as the slope of demand.³ Marginal costs are given by the horizontal line $MC = P_i$, so we get x_i where this line hits marginal revenue. The retailer marks up the price of the good to p_i , and the retailer's profits equal the area of light shading. The area of darker shading will be explained momentarily.

So how should M set P_i ? If she sets the wholesale price at P_i , the retailer will purchase $(A - P_i)/2B_i$. Hence her profits will be $(P_i - c)(A - P_i)/2B_i$. In figure 9.3, this is the more darkly shaded region. She picks P_i to maximize these profits, which leads to $P_i = (A + c)/2$, hence (see

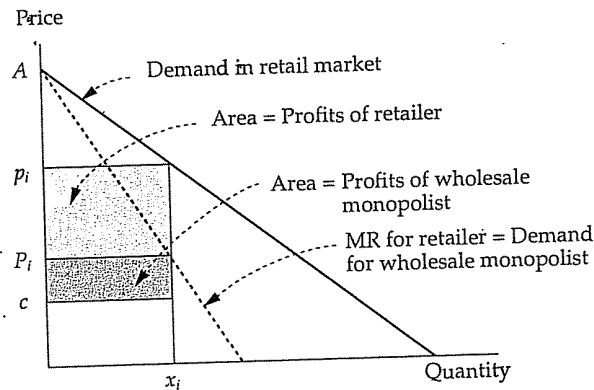


Figure 9.3. Retailer demand and mark up at price P_i .

Given the manufacturer charges retailer i the per-unit price P_i , the retailer purchases $(A - P_i)/2B_i = x_i$ units and marks the price up to $(A + P_i)/2 = p_i$. The more lightly shaded region gives the retailer's profits, and the more darkly shaded region gives the manufacturer's profits.

³ Does this square with the geometrical formula given in figure 9.2?

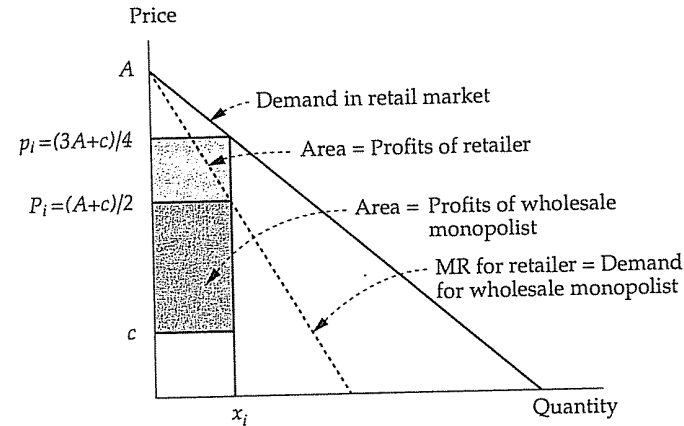


Figure 9.4. The optimal value of P_i is $(A + c)/2$.

figure 9.4) $p_i = (3A + c)/4$, $x_i = (A - c)/4B_i$, $\pi_i = (A - c)^2/(16B_i)$, and $\Pi_i = (A - c)^2/(8B_i)$, where π_i is the retailer's profits and Π_i is M 's profits from this particular part of her operations.

Note two things here. First, we assumed that M could charge a different unit price to the various retailers. Because the intercept A is the same in all the regions, she doesn't use that power of discrimination; she charges $P_i = (A + c)/2$ regardless of i . Second, if M engaged directly in the retail business in region i , she would set the retail price p_i to maximize her profits $(p_i - c)(A - p_i)/B_i$, which gives $p_i = (A + c)/2$. That is, if M sold the good directly, more would be sold at a lower price than when M must go through a (monopoly) retailer.

Now consider nonlinear prices. In particular, imagine that M , recognizing that she is the only person who can produce poiuyts, decides to charge the retailers a per unit price P_i and a franchise fee F_i . Of course, she can't compel the retailers to pay this fee, but she can (and does) insist that they pay the fee if they want to purchase any poiuyts from her. Note that we've subscribed both the per unit price and the franchise fee. We'll assume for now that she can perfectly discriminate.

So what are the optimal P_i and F_i to set? Retailer i will face the total cost function $TC(x_i) = F_i + P_i x_i$ (for $x_i > 0$), so this retailer's marginal costs will be P_i , and, assuming he buys any at all, he will purchase $(A - P_i)/2B_i$, mark them up in price to $(A + P_i)/2$, and realize gross profits (gross of the franchise fee) of $(A - P_i)^2/4B_i$, just as before. In fact, he'll do all this if his net profits $(A - P_i)^2/4B_i - F_i$ are nonnegative. (We'll assume

that if he just breaks even, he'll still buy and sell poiuyts. If this is wrong, then the analysis to follow changes a bit; F_i should be set just a bit below what we derive below, leaving retailer i with a small net profit, so that he'll stay in business.) So, pretty clearly, M wants to set the franchise fee F_i at $(A - P_i)^2/4B_i$, leaving retailer i with precisely zero profits. Of course, as she raises P_i , she decreases the franchise fee she can set. But she'll make more on the per unit sales. Where is the optimum? It comes as the solution to

$$\max_{P_i} (P_i - c) \left(\frac{A - P_i}{2B_i} \right) + \frac{(A - P_i)^2}{4B_i},$$

where the first term represents her profit from per unit sales, and the second is the franchise fee she'll collect. To find the optimum, we set the derivative of the expression above equal to zero, which is

$$\frac{A - P_i}{2B_i} - \frac{P_i - c}{2B_i} - \frac{2(A - P_i)}{4B_i} = 0.$$

The first and third terms cancel out, and we are left with $(P_i - c)/2B_i = 0$, or $P_i = c$.

This simple answer may surprise you a bit at first, but it really is very intuitive. M is going to set F_i at retailer i 's gross profits. Thus, in the end, M is going to wind up with profits equal to the *two* shaded areas in figure 9.3, so P_i should be set to maximize those two areas together. But the two areas together are just the profits that M would make charging p_i in the retail market, if M marketed the product directly. We know that the optimal p_i for direct retailing by M is $(A + c)/2$. So M wants to set P_i so that retailer i will charge that price, which is what $P_i = c$ accomplishes.

Note that while P_i is independent of i , F_i is not. If B_i is larger than B_j , market i is *smaller* than market j . Thus retailer i will make smaller gross profits than will retailer j , and the franchise fee set for i can't be set as large as the franchise fee for j . Our manufacturer M is certainly using her powers of discrimination here. This, by the way, is not an outrageous use of the powers of discrimination. One generally thinks that price discrimination won't work in cases where the good can be resold by one customer who is charged a lower price to another who would be charged a higher price. But the franchise to sell a product is not something that can be resold so easily (or, at least, the contract that gives the franchise can be written so that the franchise would be hard to resell). And it is not altogether ridiculous to think that, say, a McDonald's

franchisee in a potentially very large market will have to pay a higher fee than a McDonald's franchisee in a potentially small market.

Nonetheless, let us consider what would happen if M were forced by law or by circumstances to charge the same franchise fee and per unit cost to all the retailers. It doesn't quite make sense in this setting, but if M is unable to tell which retailers have which B_i but knows the distribution of B_i within the population, this restriction on her behavior would be natural; she would have to charge the same (nonlinear) prices to all her customers, and let them, by their actions, identify themselves in terms of their characteristics. That is to say, the sort of analysis we will take on at this point is the appropriate form of analysis for cases where a monopoly uses nonlinear prices to discriminate among a heterogeneous population, but where the monopoly is unable to tell a priori which sort of customer is any given customer.

We'll be very specific at this point. Suppose that $A = 12$, $c = 0$, and there are eight retailers: three have $B_i = 9$ and five have $B_i = 6$.⁴ How should M set P and F (without subscripts) now? In particular, will $P = c = 0$ be the solution?

Suppose that P is set at zero. Then the gross profits of the three retailers with $B_i = 9$ are $A^2/4B_i = 144/36 = 4$, while the five with $B_i = 6$ will have gross profits of 6. Our manufacturer will consider only two possible values for F . She can set $F = 4$, so that all eight retailers will sign up. This will net her profits of 32 (and will leave 2 for each of the five retailers in the relatively large markets). Or she can set $F = 6$. This will cause the three retailers with the smaller markets to refuse to buy franchises, but will net 6 each from the five larger markets, for a net of 30. The point is that $F < 4$ is silly, since raising F up to 4 will simply increase her profits. And once $F > 4$, she may as well push F all the way up to 6. So, with $P = 0$, the best thing to do is to set $F = 4$, giving her profits of 32.

Let us try $P = 1$ next. Now the gross profits of the two types of retailers will be 3.3611, on unit sales of .6111 for the three smaller retailers, and 5.042, on unit sales of .91666 for the five larger retailers. (Be sure that you can replicate these numbers.) If M sets $F = 5.041$, she'll get the five larger retailers to participate, netting $5.041 + .91666 = 5.957666$ from each (the sum of the franchise fee plus the net of unit sales), for a total profit of 29.78 (approximately). No surprise there — we know that if she is going to deal with the five larger retailers only, she can (without discriminating) charge the best combination of F and P for them, namely $F = 6$ and

⁴ If $c = 0$ bothers you, set $A = 13$ and $c = 1$, and you'll get the same answers.

$P = 0$. So setting $P = 1$ and then charging whatever F will get these five only must be suboptimal.

But she could also try $F = 3.3611$ in conjunction with $P = 1$. Then all eight retailers will deal with her. Her net profits will be approximately

$$8 \times 3.3611 + 5 \times 1 \times .91666 + 3 \times 1 \times .61111 = 33.3.$$

Note that this is eight franchise fees, plus .91666 sold to five clients at a price of 1 apiece, plus .6111 sold to three clients at a price of 1 apiece. And, sure enough, this gives greater profits than the best scheme with $P = 0$. What is happening here is that, since M can't extract all the profits from the five larger retailers (without losing the three smaller ones), she is forced back in the direction of the scheme where she can't charge a franchise fee at all, where P is not zero.

So the optimal P won't be 0. What will it be? You can work this out if you wish, although you may find it more challenging to try problem 3, where you are asked to work out a slightly more complex version of this problem.

9.5. Monopoly power?

In all of the previous discussion, it was assumed that the monopoly could name its price (or, equivalently, the quantity that it would sell) or could make a take-or-leave offer to potential consumers. You might wonder just why this is supposed to be true. Consider, for example, our "fortunate" monopoly, which can make a different take-or-leave offer to every consumer and which knows the characteristics of each consumer. Suppose it makes this take-or-leave offer to consumer X , who says "no thanks," and then comes back with a take-or-leave offer of her own. For example, consumer X might offer to take n units off the hands of the monopoly, at (being generous) \$1 above the monopoly's marginal cost for producing the n units. Wouldn't the monopoly take this deal, if it really believed it was the best deal it could get? And, if so, how do we determine who, in this sort of situation, does have the bargaining power? Why did we assume implicitly that the monopoly had all this power (which we most certainly did when we said that consumers were price takers)?⁵

⁵ For those of you who remember back to a principles or intermediate micro course, another way to think of this is to consider that the monopoly dealing separately with a single consumer is in a position of *bilateral monopoly* which, you will recall, is meant to have an indeterminate solution.

9.5. Monopoly power?

Standard stories, if given at all, get very fuzzy at this point. Hands start to wave, hems give way to haws, and way is quickly given to the next subject. The handwaving typically involves something about how there is one monopoly and many consumers, and it can do without any one of them, but they can't do without it. Somehow (and this is never made precise), the monopoly's advantage in numbers gives it a credibility about setting and sticking to a price or in sticking to its take-or-leave offer.

An important variation on this basic theme explores the extent to which the monopoly is its own worst enemy when it comes to sticking credibly to a price. Imagine a monopoly that is able to produce a particular good at a cost of \$1 per good. Demand for this good is given by the demand function $D(p) = 25 - p$. If you solve the monopoly's problem, you'll find that the optimal solution is for it to set $p = 13$ and to produce 12 units of the good. So suppose it does this — it sells the 12 units at \$13 apiece. Once those twelve units are sold, why doesn't our monopoly decide to sell a few more at, say, \$8 apiece? At \$8, there is unsatisfied demand, and our monopoly will net \$7 for every additional unit it sells. After selling what it can at \$8 apiece, why not cut the price again, to say, \$4, and so on?

The problem with this is that if you are a consumer who values the good highly (you are willing to pay \$13, or even \$24, if it comes to that) and if you think that the monopoly is going to cut the price after it sells off a few at a high price, then you will wish to wait for the price to drop. If all consumers rationally anticipate the drop in price, then no one will buy at \$13.

Note well that the monopoly itself is the real problem here. Once it makes some sales at \$13, if it makes any (and even if not), it is in its own best interests *ex post* to drop prices and sell more. It won't be in its own best interests *ex ante* to be perceived in this fashion. But if everyone understands that, *ex post*, this will be in its interests and this is what it will do, then there isn't much it can do to convince people that it won't drop price.

The problem we are alluding to here is variously known in the literature as the Coase conjecture and the problem of a durable goods monopoly. (The relevance of durability comes from the notions that (a) the market isn't continually regenerating itself, as is, for example, the market in wheat, and (b) if the good is durable, waiting for price to drop won't hurt a consumer too much.) The Coase conjecture, stated a bit more baldly than normal, is that a monopoly doesn't have monopoly power. It won't be able to keep itself from dropping the price *ex post*, and so everyone expects to see prices drop. But then no one will buy at a high price. And (so the

argument goes, when worked out fairly rigorously) a monopoly can't get anyone to buy above the price at which it is just willing to sell — its marginal costs.⁶

Once again, the problem is one of credibility. Before we were asking how the monopoly credibly maintains that it won't haggle with an individual customer. Now, how does the monopoly credibly maintain that it won't drop the price it charges as time passes?

It may well be that the monopoly can make credible commitments not to haggle and not to lower its price. We don't need to know why and how such commitments can be made credibly if we are testing the theory empirically; if, in a particular market, a monopoly acts the way the theory suggests, then we suppose that somehow or other the firm has this credibility. But a more precise analysis of why and how it has this sort of credibility would help refine any empirical investigation that we conduct.

What would a more precise analysis look like? Let us give two rough examples:

(1) Certain forms of art, lithographs and castings, can be made cheaply in multiple copies. So a common practice of artists who make lithographs and castings is to "break the mold" after making a given number of prints/casts: This practice is clearly directed towards the durable goods problem; the artist would otherwise always have an incentive to produce and sell more (down to the marginal cost of production, which is very low). More generally, a monopoly seeking to commit to produce a certain quantity and no more might try to commit itself to a production technology where the marginal cost of production rises precipitously past the level it wishes to produce. With such a technology, the monopoly could credibly refuse to haggle on grounds that the marginal cost of producing an additional unit (beyond the monopoly quantity) is prohibitive.

(2) The monopoly, in a fit of "largesse," could offer all its customers "most favored customer" guarantees. Such a guarantee might state, for example, that if the monopoly ever sells the good in question at a price below that which a particular customer pays, then the customer is entitled to a rebate of the difference (with accrued interest, say). The effect of this is dramatic. To conduct a precise analysis takes techniques that we don't have at our disposal yet, but the following assertions are likely to appeal to your

⁶ More precisely, the extent to which a monopoly can exploit its monopoly power depends on two factors: the extent to which the monopoly can limit its own ability to drop the price it quotes and the impatience of customers. If, for example, the monopoly cannot change its price more than once a month, and if some customers are unwilling to wait a month for a somewhat lower price, then the monopoly can "exploit" those customers.

intuition. If some single customer attempts to haggle with the monopoly, the monopoly will point out the dire consequences to it of lowering the price for even one customer (namely, it means lowering the price for all). Similarly, lowering the price as time passes is no longer attractive. Note that insofar as the monopoly's credibility problems prevent it from otherwise charging monopoly prices, it is in the interests of customers to have a *general* prohibition against such guarantees. But each customer sees it in her personal interest to take such a guarantee if offered, so we can imagine that such guarantees might be given and taken, if conditions would permit them to be enforced effectively.^b For such a guarantee to be enforceable, each customer would have to be able to observe all the deals the monopoly makes. It is, therefore, in the interest of monopoly and against the collective interests of customers to arrange matters so that all sales are a matter of public record, so that the guarantees will be effective.

These are only two examples. There are other sources of the type of credibility that the monopoly would like to have. In particular, in chapter 14 we will explore models in which the monopoly's *reputation* could provide the backbone needed to name and stick to a price.

Our point for now is primarily methodological: In the two examples sketched (and, it will turn out, when we look at models of reputation), we address the monopoly's ability to commit to a price by examining the institutions of production and exchange. As long as we avoid these institutional details, it is hard to say much about whether a monopoly can muster the credibility required. More generally, many of the questions posed (but not answered) in this chapter arise from the lack of specification of the institutional environment. To investigate these questions, it would seem useful to get more deeply involved with the details of the monopoly's environment. We will pick up on this theme even more strongly in the next chapter.

9.6. Bibliographic notes

The objective of this chapter has been to raise some issues that we hope will lead you to conclude that rather a lot is interesting to resolve concerning monopoly; certainly there is more to the subject than recitation of $MR = MC$. Some of these issues will be resolved when we get to the appropriate tools in later parts of the book. But for the most part, you will have to read elsewhere for the resolution of these questions. An

^b That is, we are seeing here a reason why it might matter that there is one monopoly and many customers; viz., the many customers may by their numbers be unable to act collectively.

excellent treatment of all manner of material that has been touched on in this chapter (and much that has not) is given by Tirole (1988). Part I of Tirole covers the basic theory of monopoly and a host of topics including the Coase conjecture, multigood monopolies, nonlinear pricing, the problem of a manufacturing monopoly selling to retailing monopolies, product selection, and advertising. His chapters 8 and 9 give treatments of entry deterrence. (You will have a much easier time with these two chapters after completing part III of this book.)

On the subject of nonlinear pricing and price discrimination, also highly recommended are Philips (1983) and Wilson (forthcoming).

References

- Philips, L. 1983. *The Economics of Price Discrimination*. Cambridge: Cambridge University Press.
- Tirole, J. 1988. *The Theory of Industrial Organization*. Cambridge, Mass.: MIT Press.
- Wilson, R. Forthcoming. *Nonlinear Pricing*. Palo Alto, Cal.: Electric Power Research Institute.

9.7. Problems

- 1. Why is the geometrical procedure for finding the value of marginal revenue at a quantity x given in figure 9.2 correct? Why do we know that a monopoly (which maximizes profits in the traditional way) will produce at larger than efficient scale in figure 9.1(a) and smaller than efficient scale in figure 9.1(b)?
- 2. What is wrong with the picture in figure 9.5, which is meant to describe the situation of a monopoly I know?
- 3. Recall that in section 9.4 we considered a large manufacturing monopoly selling to individual retailing monopolies. There were eight retailers, three of whom face demand of the form $p = 12 - 9x$, and five of whom face demand of the form $p = 12 - 6x$. Well, it turns out that there are actually ten retailers — the eight described above and two more who face demand of the form $p = 12 - 2x$. Costs of production and sales are zero. The retailers are price takers in their dealings with the manufacturer. The manufacturer can set a fixed fee F and a per unit charge P ; the retailers take these as given and decide (a) whether to participate at all, and (b) if so, how many units to purchase. The fixed fee and per unit charge must be the same for all the retailers; no discrimination is allowed.

9.7. Problems

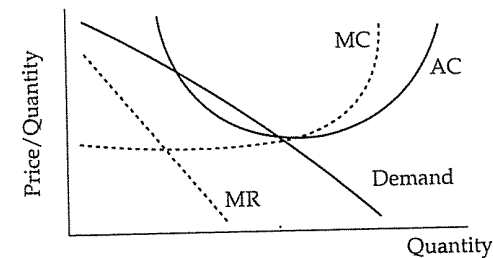


Figure 9.5. Picture for problem 2.

- (a) What is the optimal fixed fee to charge if P is set at zero?
 - (b) What is the optimal fixed fee to charge if $P = 1$?
 - (c) What is the optimal fixed fee and per-unit charge?
- 4. In problem 3(c), you found the optimal nondiscriminatory fixed fee-per unit charge pricing scheme for the manufacturer to employ. Now consider general nondiscriminatory schemes. Imagine that the manufacturer is able to publish any sort of price scheme she wants. Being formal about it, the manufacturer picks a function $\Phi : (0, \infty) \rightarrow [0, \infty]$ with the following interpretation: A retailer can purchase quantity x from the manufacturer at a total cost of $\Phi(x)$. We allow $\Phi(x) = \infty$ for some quantities x , with the interpretation that the manufacturer is unwilling to sell that quantity. The manufacturer is able to control resale of this good, so each retailer must live with the scheme Φ ; given Φ , each retailer chooses that level x to buy such that its revenues selling x less its costs $\Phi(x)$ are as large as possible. But the manufacturer is unable to discriminate among the retailers; the manufacturer must set a single cost schedule Φ and allow the different retailers each to choose his own preferred point.
- (a) Prove that the best the manufacturer can do with such a general nondiscriminatory scheme is strictly worse than what she can do if she is allowed to discriminate (and use general schemes).
 - (b) To what extent does your proof in part (a) depend on the exact numbers and parameterization in problem 3? Specifically, suppose that we looked at an example in which there were a number of different retailers, all of whom faced demand with the same B but with different A_i .

Would your proof work in this case? And what if both the A_i and B_i varied? Would your proof still work in all cases? Would it only work in some? If the latter, can you characterize precisely those circumstances in which a nondiscriminatory scheme can do just as well as the optimal discriminatory scheme?

(c) Go back to the parameterization of problem 3. What is the best general nondiscriminatory scheme from the point of view of the manufacturer? (This is difficult. You will be in better shape to tackle this part of this problem after finishing chapter 18.)

- 5. Recall from chapter 8 all the problems about the pillip industry. In particular, recall that you solved in problem 3 for the market equilibria (in pillip and kapitose) when the kapitose supply function was given by $q = K/200$ and the demand for pillip was given by $D(p) = 750 - 150p$. Now consider what will happen in these two industries if the supply of kapitose is controlled by a monopoly whose marginal costs for producing K units of kapitose are $K/200$. (You may well need to resort to numerical solutions.)

We did nothing in this chapter in the way of deriving testable propositions from the standard model. Just so you don't get away without at least a taste of this:

- 6. Consider a monopoly that faces a downward sloping demand curve $X = D(P)$ and constant unit costs c . We are interested in how this monopoly will adjust the price it charges as c changes.

(a) If demand is linear $X = A - P$, show that less than the full change in costs is passed on to the customers. (That is, if we think of $p(c)$ as the monopoly price, given as a function of cost c , then $dp(c)/dc < 1$.)

(b) Suppose that demand takes the form $X = P^{-\alpha}$ for $\alpha > 1$. Show that the monopoly passes on more than the full cost increase to consumers. (That is, $dp(c)/dc > 1$.)

(c) For which demand functions (if any) will the monopoly precisely pass on to consumers any cost increase? (That is, for which demand functions will $dp(c)/dc = 1$?) You should give demand functions for which this works at every level of c . (Hint: If you begin by drawing the obvious picture in part [a], it may help you see what to do in part [c].)

Since we introduced linear demand in the last two chapters, it might be a good idea to say something about where it comes from. So:

- 7. In an economy that I know, there are many consumers. One of the

goods consumed in this economy is phiffle, a nonnarcotic relaxant. It turns out (for reasons that needn't concern you) that we can represent the preferences of every consumer in this economy as follows: If the consumer consumes x units of phiffle and has $\$z$ left to spend on all other commodities, whose prices are held fixed throughout this problem, the consumer's utility has the form

$$u_i(x) + k_i z,$$

where the subscript i refers to the consumer's index, u_i is a given function, and k_i is a given positive constant. This consumer must consume a nonnegative amount of phiffle and cannot spend more money on phiffle than he has in total, but is otherwise unconstrained.

(a) Suppose the consumer has $\$w$ to spend in total, and the price of phiffle is $\$p$ per unit. Set up, as a constrained maximization problem, the problem of the consumer (who is a price taker) in deciding how much phiffle to consume. Give first-order conditions and (if relevant) complementary slackness conditions.

(b) Suppose in this economy there are ten consumers. Each of these consumers has $u_i(x) = \sqrt{x}$. Three of the consumers have $k_i = 4$; four have $k_i = 3$; and three have $k_i = 2$. Each consumer has $\$1,000$ to spend. What is the market demand curve for phiffle in this case?

(c) Continue to suppose the data from part (b). Imagine that the production of phiffle was controlled by a monopoly with rising marginal costs given by $MC(x) = x$. What will be the price and quantity delivered by this monopoly?

(d) Now we suppose in this economy there are 100 consumers. Each of these consumers has $k_i = 1$, and each has $u_i(x) = x - c_i x^2$ for $x \leq 1/(2c_i)$ and $u_i(x) = 1/(4c_i)$ for $x \geq 1/(2c_i)$. Moreover, the 100 consumers are indexed by $i = 1, \dots, 100$ and for consumer number i , $c_i = i$. All the consumers in this industry have initial wealth of (at least) 2. What is the market demand curve for phiffle in this case?

(e) Suppose in a somewhat larger economy there are 10,000,000 consumers. This economy is broken into four regions. In region 1, are 4,000,000 consumers; in region 2 are 3,000,000; in region 3 are 2,000,000; and in region 4 are 1,000,000. Inside each region, the consumers are a scaled-up replica of the 100 consumer economy of part (d). That is, in region 2 (say) are 30,000 consumers with $c_i = 1$, 30,000 with $c_i = 2$, etc. What are the demand functions for phiffle in each of these regions?

- 8. Suppose a monopoly facing the sort of credibility problem sketched in section 9.5 decided to be generous to its customers and offer each one a “most favored customer” contract as outlined there. Create and analyze a formal model that supports some (or, if you can, all) of the intuitive assertions we made. You may ignore the time value of money, to keep the analysis relatively simple. You may wish to return to continue your analysis after you have completed part III of the book.

In the next three problems, you are introduced to ways of modeling demand for differentiated goods and to the problem of a monopoly deciding how many “varieties” of a good to provide. We look at a particularly simple case of this problem; see Tirole (1988, chap. 2) for a much more complete discussion.

- 9. Imagine a single consumer with the following very strange pattern of demand for a particular good. This good, which we will call a griffle, can come in any number of varieties. Specifically, any griffle is of a type t for some $t \in [0, 1]$. Our consumer is interested in purchasing at most one and precisely one griffle; he will do so if there is a griffle whose price and type meet a hurdle that he sets. His preferences and demand behavior for griffles are characterized by a function $r : [0, 1] \rightarrow (0, \infty)$ as follows. Suppose N types of griffles are for sale, types $t_1, t_2, \dots, t_N$. Suppose that $p(t_n)$ is the price of a griffle of type t_n . Then if for each n , $p(t_n) > r(t_n)$, our consumer will not purchase any griffle. On the other hand, if for some t_n , $p(t_n) \leq r(t_n)$, then our consumer will purchase precisely one unit of some type n that maximizes the difference $r(t_n) - p(t_n)$. Note well: If more than one type n maximizes this difference, our consumer will purchase precisely one unit of some one of the maximands; it is indeterminate which he will purchase.^c

(a) Suppose that our consumer’s demand behavior is characterized by the function $r(t) = r_0 + k|t - t_0|$ for some constants r_0 , k , and $t_0 \in [0, 1]$. Being very specific, suppose that $r_0 = 5$, $k = 1$, and $t_0 = .6$. Suppose three types of griffle are being sold: types .3, .5, and .7 at prices 3, 3.3, and 4, respectively. What will our consumer do (in terms of purchasing griffle) in these circumstances?

(b) Fix the prices of types .3 and .7 at 3 and 4, respectively. Draw this

^c You may worry that this sort of demand doesn’t conform to our models of consumer preference and demand in chapter 2. But except for an unhappy lack of continuity, this demand behavior can be rationalized by preferences that are representable with a utility function. If you like a real challenge, give one such utility function.

consumer’s demand curve for griffle of type .5 as a function of the price of this type of griffle.

You may wonder how to interpret these types of good. One interpretation appropriate for the particular function $r(t)$ given in problem 9 is that there is really only one type of griffle, but the different types refer to locations at which griffle is sold. More precisely, imagine that consumers live along a 100-mile-long highway. Our consumer lives at milepost 60. Our consumer is willing to pay 5 for a unit of griffle, as long as that griffle is delivered to her door. But there is no griffle delivery service. Instead, there are stores along the highway where griffle is sold. Specifically, there are stores at mileposts 30, 50, and 70. It costs our consumer .05 to drive one mile, and she counts this against the purchase cost of griffle. So if griffle is being sold for 3 at the store at milepost 50, she figures this means a roundtrip of 20 miles and an “effective” price to her of $3 + (.05)(20) = 4$. She is willing to buy a unit of griffle if it is sold somewhere at an effective price of 5 or less, and if it is sold at this reservation price or less, she buys it from wherever the effective price is lowest.

- 10. Now suppose there are many consumers, all of whom have demand of the sort described in problem 9, but with different values of r_0 , k , and t_0 . We imagine, in fact, there is a “mass” of consumers given by a density function $\phi(r, k, t)$ defined on the space $[0, 10] \times [0, 10] \times [0, 1]$. That is, for any subset $A \subseteq [0, 10] \times [0, 10] \times [0, 1]$, the number of consumers whose characteristics (r_0, k, t_0) lie within the set A is $\int_A \phi(r, k, t) dr dk dt$.^d If ϕ is a uniform density, what does the aggregate demand function for griffle of type .5 look like, if griffle of types .3 and .7 are for sale at prices 3 and 4, respectively. (If you find this too hard, assume that all consumers have $r_0 = 5$, $k = 1$, and t_0 distributed uniformly on the interval $[0, 1]$.)

- 11. Assume consumers are distributed according to a uniform density ϕ as in problem 10. (If that is too difficult, answer both parts of this question assuming that only t_0 is distributed among the population, with uniform distribution on $[0, 1]$.) Suppose a monopoly manufactures and sells griffle. This monopoly has a constant marginal cost technology, with marginal cost equal to 1.

(a) If the monopoly may sell only one variety of griffle, what variety of griffle will the monopoly sell and at what price?

(b) Suppose the monopoly may sell as many varieties of griffle as it chooses, but it must pay a fixed cost of 1 for every variety of griffle it

^d If you know measure theory, generalize suitably.

chooses to offer. Suppose that the total mass of consumers is 100; that is, $\phi(r, k, t) \equiv 1$ for all $(r, k, t) \in [0, 10] \times [0, 10] \times [0, 1]$. How many varieties of griffle will the monopoly offer? What varieties will be offered? What price will be charged for each?

chapter ten

Imperfect competition

We move next to models of imperfect competition. Firms in these models have rivals, and the actions of their rivals affect how well they do. But at the same time firms are not price takers; when they optimize, they take into account how their actions affect the prices they face both directly and, through possible reactions of their rivals, indirectly.

We will give only a bare introduction to the subject. As in chapter 9, we will cover the rudiments, in this case the classic models of duopoly with undifferentiated goods: Cournot, von Stackelberg, Bertrand, and kinked demand. Then we will discuss informally and briefly some of the many important issues that impinge on the basic models. (The problems present the opportunity to move beyond informal discussion.)

10.1. The classic models of duopoly

We imagine an industry with two vendors/firms, each producing and selling a single good. Consumers don't care from which firm they purchase the good. Demand, as in the theories of perfect competition and monopoly, is given by a demand curve. We'll take a very concrete and simple case of linear demand: $X = A - P$, where P is price, X is total quantity demanded, and A is a constant. We'll assume that the two firms are identical, with constant marginal costs of production k and no fixed costs. (We assume that $A > k$.)

Just to set the stage, we recall the competitive and monopolistic outcomes in this setting. In a competitive industry, price (in the long run) would equal minimum average cost, which in this very simple setting is marginal cost k . So quantity sold would be $X_c = A - k$. If the industry were a monopoly, the monopolist would equate marginal cost k with marginal revenue $A - 2X$, giving equilibrium price $P_m = (A + k)/2$ and equilibrium quantity $X_m = (A - k)/2$.

What differentiates the various models of duopoly are the *conjectures* that each industry participant makes concerning the actions and reactions

of its rival. In the typical model, one makes assumptions about those conjectures and then finds the corresponding equilibrium outcome, where an equilibrium is a point at which neither firm in the pursuit of maximal profits wishes to change its own action, given the (re)actions that it *supposes* others will take if it changes its own action.

Cournot equilibrium

In the first model we consider, each firm has *Cournot conjectures* about the actions of its rival: Each of the two firms assumes that the other firm will act in a way to keep fixed the quantity that it sells.

If we suppose that each firm has these conjectures about the other, we can compute what is called the *reaction function* of each to the quantity choice of the other. Suppose that firm 2 has output at level x_2 . Firm 1, since it has Cournot conjectures, supposes that firm 2 will not change this level of output. So firm 1 conjectures that as a function of its level of output x_1 , price is $A - x_2 - x_1$. The profits of firm 1 as a function of x_1 will be $(A - x_2 - x_1)x_1 - kx_1$. This is maximized (as long as $x_2 \leq A - k$) at

$$x_1^*(x_2) = (A - k - x_2)/2.$$

(To see this, simply differentiate the profit function of firm 1.) This reaction function of firm 1 to firm 2 says how firm 1 reacts (optimally) to the quantity choice of firm 2, based on firm 1's Cournot conjectures about firm 2. Similarly, firm 2's reaction function to firm 1's choice of quantity x_1 is given by

$$x_2^*(x_1) = (A - k - x_1)/2.$$

We are at an *equilibrium* in this industry when neither firm wants to change what it is doing, given how it believes the other firm will react to any change. Here the assumed reaction is to keep the same level of output. So we are at an equilibrium when we are at quantity levels $(\hat{x}_1, \hat{x}_2)$, which are on the two (optimal) reaction functions. That is, $(\hat{x}_1, \hat{x}_2)$ should satisfy $\hat{x}_1 = x_1^*(\hat{x}_2)$ and $\hat{x}_2 = x_2^*(\hat{x}_1)$. Since we have these two reaction functions, we can solve these two equations (in two unknowns) and find that the unique solution is $\hat{x}_1 = \hat{x}_2 = (A - k)/3$. Alternatively, we can graph the two reaction functions in (x_1, x_2) space and find this intersection point graphically; this is done for you in figures 10.1 and 10.2. In figure 10.1, we graph the reaction function of firm 1 to firm 2, with iso-profit curves

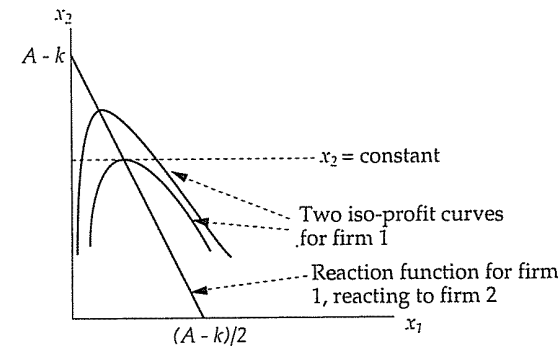


Figure 10.1. A Cournot reaction function and the iso-profit curves from which it is built.

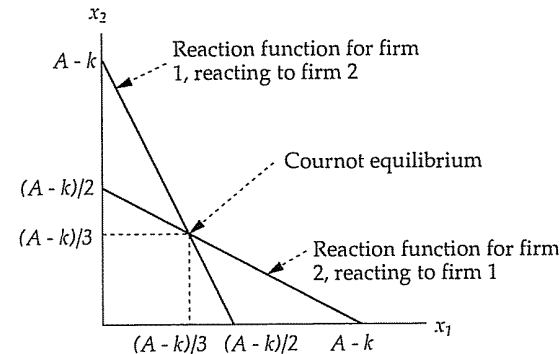


Figure 10.2. The Cournot equilibrium.

for firm 1.¹ Note that the reaction function for firm 1 is the locus of points where its iso-profit curves are tangent to lines of the form $x_2 = \text{constant}$. In figure 10.2, we superimpose the reaction functions of the two firms. Where they intersect is the Cournot equilibrium.

At this *Cournot equilibrium*, total output is the sum of the two equilibrium quantity choices, or $X_C = 2(A - k)/3$, and price is $P_C = A - X_C = A/3 + 2k/3$. Note that the Cournot equilibrium price is higher than the competitive price but lower than the monopoly price. Prices aren't

¹ An iso-profit curve for firm 1 is a locus of points (x_1, x_2) that give the same level of profit to firm 1; that is, for a given constant C , the C -iso-profit curve is the set of points such that $(A - x_1 - x_2 - k)x_1 = C$.

competitive, because each firm takes into account that it depresses price when it raises output. (That is, firms are not price takers.) But firms, by raising quantity, don't feel the full brunt of the depressed price; their competitor feels some of this as well. Hence the equilibrium is at a lower price and higher quantity than would be the case for a monopoly.

In the analysis above, we took as the decision variable the quantity chosen by each firm. We could, with greater algebraic difficulty, have made price the decision variable. At the risk of confusing matters, let us go through this exercise: Suppose firm 2 is producing a quantity x_2 . Firm 1 can choose the optimal equilibrium price P for itself under the Cournot conjectures that x_2 will remain the output of firm 2. Of course, not every price P is available to firm 1; it can't have a price $P > A - x_2$, since such a price would be inconsistent with firm 2 selling x_2 . And, we can presume, it won't want to choose a price P lower than zero. In fact, it won't want to choose a price P lower than k , since its average costs will be k . So suppose it chooses a price P from $[k, A - x_2]$. Under its Cournot conjectures, it will sell a quantity $x_1 = A - x_2 - P$ for a profit of $(P - k)(A - x_2 - P)$. The optimal choice of P as a function of x_2 is $P^*(x_2) = (A - x_2 + k)/2$. Similarly, as a function of x_1 , firm 2 will want the market price to be $P^*(x_1) = (A - x_1 + k)/2$. We solve for the equilibrium with the conditions that at the equilibrium price and quantities the price that each side desires should be equal to each other and to the equilibrium price. That is,

$$P^*(\hat{x}_2) = P^*(\hat{x}_1) = A - \hat{x}_1 - \hat{x}_2, \text{ or} \\ (A - \hat{x}_2 + k)/2 = (A - \hat{x}_1 + k)/2 = A - \hat{x}_1 - \hat{x}_2,$$

which simplifies to the equilibrium we computed before: $\hat{x}_1 = \hat{x}_2 = (A - k)/3$. Why go through this algebra when we already had the answer? We wish to stress that to say that firms have Cournot conjectures is not to say they take quantity (and not price) to be their decision variable. Rather, Cournot conjectures are conjectures about the response of the other firm, namely that the other firm will (re)act in a way to keep the quantity that it sells fixed.^a

Von Stackelberg equilibrium

Continue to suppose that firm 2 has Cournot conjectures about the actions of firm 1, so that firm 2's reaction as a function of the quantity choice x_1 of firm 1 is $x_2^*(x_1) = (A - x_1 - k)/2$. But suppose that firm 1 does not have Cournot conjectures. Instead, firm 1 believes (correctly, as

^a Although it will seem cryptic to most readers, a further comment is in order for those of you who have read about Cournot equilibrium elsewhere. The statements just made do not contradict an assertion we will make at the end of this chapter and more coherently in chapter 12, namely that one reason to suppose that firms have Cournot conjectures is that quantity output is the decision variable of each firm. For those of you to whom this is hopelessly cryptic, wait until the end of chapter 12 before panicking.

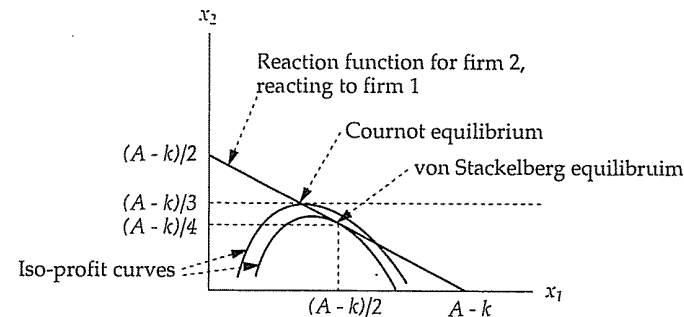


Figure 10.3. The von Stackelberg equilibrium.

it turns out) that firm 2 will react to firm 1's choice of quantity in Cournot fashion. That is, firm 1 believes that firm 2 will be influenced by x_1 ; firm 2 will set $x_2 = x_2^*(x_1)$. In this case, firm 1 selects x_1 to

$$\max_{x_1} [(A - x_1 - x_2^*(x_1))x_1 - kx_1].$$

Substituting for $x_2^*(x_1)$ and differentiating with respect to x_1 , this yields an optimal quantity choice for firm 1 of $\hat{x}_1 = (A - k)/2$, which in turn leads to a choice by firm 2 of $\hat{x}_2 = x_2^*((A - k)/2) = (A - k)/4$, total quantity of $3(A - k)/4$, and equilibrium price of $A/4 + 3k/4$.

This is known as the *von Stackelberg* model of duopoly, with firm 1 the so-called *von Stackelberg leader* and firm 2 the *follower*. To be more precise, we should call this the Cournot-von Stackelberg model, since it is assumed that the follower has Cournot conjectures, and the leader knows (and uses) this fact.

The "picture" for this equilibrium is given in figure 10.3. We see here the reaction function of firm 2, and iso-profit curves for firm 1. Firm 1 is choosing from the reaction function of firm 2 that point which maximizes its own profits.

Two von Stackelberg leaders and consistency of conjectures at the equilibrium

Suppose that both firms had the conjectures of firm 1 above. That is, firm 1 conjectures that firm 2 will respond to x_1 with $x_2^*(x_1)$ and firm 2 conjectures that firm 1 will respond to x_2 with $x_1^*(x_2)$. What would be the equilibrium in this case?

Presumably, the only possibility for an equilibrium is where both firms produce $(A - k)/2$. But is this an equilibrium? If firm 1 sees firm 2 producing $(A - k)/2$ when it is producing this quantity itself, it will surely recognize that its conjectures about firm 2 are false, and similarly for firm 2. Because of this, we would not call this an equilibrium. In general, we will say that we have an equilibrium when two conditions are met: Neither firm, given its conjectures, wishes to change what it is doing; and the equilibrium actions of the two firms are consistent with the conjectures that each firm is supposed to hold. In the Cournot and the single-leader von Stackelberg stories, the equilibrium quantities (and prices) were consistent with the firms' conjectures. In the case of Cournot, this is by default; each firm conjectures that the other won't change its quantity, which is consistent with any pair of quantities for the two firms. In the case of von Stackelberg, there is a bit more to this: Firm 2's conjectures about firm 1 (that firm 1 will not change its quantity) hold by default; and firm 1's conjectures about firm 2 hold because of firm 2's conjectures and corresponding reaction. But in the case of two von Stackelberg leaders, there is no equilibrium. The only point at which firms' conjectures about each other are consistent with the situation is where each firm produces $(A - k)/3$. And at that point, both firms, given their conjectures, wish to increase their output.

We will not stress this consistency condition in what follows except in one place. But you should check along the way that the equilibria we describe do give situations where each firm's conjectures about the other are not disconfirmed by the equilibrium activity levels.

Bertrand equilibrium

The two firms are said to have Bertrand conjectures if each believes that the other will not change the price that it is quoting. Now this poses something of a quandary. Suppose the equilibrium price of a good is, say, \$10. We typically assume that this means that both firms (in a duopoly) charge \$10. Put yourself in the place of one of the two firms and imagine that you (a) have Bertrand conjectures and (b) contemplate lowering your price to \$9 (or otherwise taking actions that would cause the market price to fall to \$9). What, precisely, does it mean to say that your rival will maintain a price of \$10?

Presumably, since the goods in question are undifferentiated, everyone would rather buy from you than from your rival, if you are really selling at \$9 and your opponent at \$10. So we will assume just this: If, in these circumstances, you contemplate lowering your price to \$9, and you anticipate that your rival will hold to a price of \$10, then you anticipate

that the whole of the market demand at \$9 will queue up outside your door clamoring to make a purchase from you. Your opponent, meanwhile, will face a queue without customers. Now, will you be willing to serve all those customers? Perhaps yes and perhaps no — it depends on how many customers are queued and at what point your marginal costs equal \$9. But, for the time being, we will assume that for some reason (perhaps legal) you are forced to sell all the demand that comes to your door.

At the same time, if in a given situation the market price is \$10, you hold Bertrand conjectures, and you contemplate raising your price, then you contemplate that all your customers will go to your rival, and you make no sales.

Now suppose that you are in an equilibrium situation, where both firms charge the same price, say \$10. And suppose that at a price of \$10, there is demand of $A - 10$. How is that demand divided between the two firms? This is quite indeterminate. Any number of divisions of market demand between the two firms is consistent with a price of \$10. So in looking for a Bertrand equilibrium, we will describe situations in terms of the prices firms are charging and the demand they are serving, where prices and demand served must be consistent with the rules: If the two firms charge unequal prices, the demand served by the low-price firm must equal all of market demand at that price (and the high-price firm gets no sales); and if the two firms charge the same price, total market demand at that price is divided in some arbitrary, but specified, way between the two firms.

Why didn't we need to worry about this in the context of Cournot and von Stackelberg equilibria? That is, why did we simply suppose that the two firms would both wish to charge the market clearing price, so that we could deal in an equilibrium in terms of the two quantity levels of output? For one thing, in any situation where both firms sell strictly positive amounts, they must be selling at the same price, at least in the model under consideration. But still we might worry about an equilibrium where one firm charged a higher price than the other and made no sales; if you wish to be very careful about things, you can go back and reassure yourself that in neither of the previous two cases would such a situation be an equilibrium.²

We could next try to construct "reaction functions" just the way we did in the subsection on Cournot conjectures, to find equilibria. But each

² Once again, for those of you who have seen the Bertrand "story" before and remember it as being less convoluted than this, bear with this convoluted story. This way of doing things will eventually lead you to a better understanding of just what is going on than will direct reference to the less convoluted story that you may have heard.

firm's reaction to a particular "situation" is a bit more complex in this case, since how one reacts to a situation in which the two are charging the identical price may depend on how demand is divided between them. It is easier in this case to try to reason directly about what sort of situation could conceivably be an equilibrium.

An equilibrium will consist of a pair of prices $(\hat{p}_1, \hat{p}_2)$ and a pair of quantities $(\hat{x}_1, \hat{x}_2)$ where:

- (a) if $\hat{p}_1 < \hat{p}_2$, then $\hat{x}_1 = A - \hat{p}_1$ and $\hat{x}_2 = 0$
- (b) if $\hat{p}_2 < \hat{p}_1$, then $\hat{x}_2 = A - \hat{p}_2$ and $\hat{x}_1 = 0$
- (c) if $\hat{p}_1 = \hat{p}_2$, then $\hat{x}_1 + \hat{x}_2 = A - \hat{p}_1$

and, most critically (d) neither firm wishes to change the situation, given that it holds to Bertrand conjectures.

Now we proceed by cases.

(1) Could there be an equilibrium where the prices are unequal and where the smaller of the two prices is more than k ? No, because in such a situation, the firm charging the higher price is selling nothing and by undercutting the price charged by its rival this firm will capture the entire market, making positive profits as long as it undercuts its rival at a price that still exceeds k .

(2) Could there be an equilibrium where the prices are unequal and where the smaller of the two prices is less than or equal to k ? No, because in such a situation the firm charging the lower price would wish to raise its price at least a bit. If the high-price firm is charging more than k , then the low-price firm can move from a loss or zero profits (which it incurs by charging a price below or equal to k) to a profit (by charging more than k and a bit less than the other firm), and if the high-price firm is charging k or less, the lower-price firm is charging strictly less than k and so is taking losses, which it can avoid by charging anything above the price charged by the high-price firm.

(3) Could there be an equilibrium where the prices are equal to each other and equal to something less than k ? No, because then one firm or the other is taking losses (we can't say which one, because we don't know whether both are making positive sales, but one at least is doing so), and that firm would do better to change to a higher price and make zero profits.

(4) Could there be an equilibrium where the prices are equal to each other and equal to something more than k ? No, because then at least one of the two firms would do better by shaving its price slightly. To see this, note that

one or the other is selling no more than half the market demand at the equilibrium price, so that firm believes that it can more than double its sales by decreasing price slightly. This causes the profit per unit sold to decline, but if it shaves its price only a bit, then the profit per unit won't fall by half, and so total profits will increase.

So the only possibility left is where $\hat{p}_1 = \hat{p}_2 = k$. And any division of market demand between the two firms at those prices is an equilibrium. This is so because each firm makes exactly zero profits in this configuration, which is the best it can do given Bertrand conjectures about its rival. If it lowers its price, it will take a loss on any unit it sells. And if it raises its price, it will make no sales, and hence continue to have zero profit. Even if it could change its share of demand (and we haven't indicated how it might conceivably do that), it has no incentive to do so, since it makes zero profit on each and every unit it might sell.

The remainder of this subsection is a bit esoteric, and even readers who religiously read through the small type may wish to avoid it on a first reading. The analysis given above assumes that a firm must serve any demand that comes to its door. If we modify that assumption, the analysis looks a bit different. To make this precise, however, we have to specify the "mechanics" of the market — what happens if one firm refuses to serve some of the demand that comes to it. The issue here is best understood by considering a case where one firm quotes a price p_1 and the other a price p_2 , for $p_1 < p_2$. Then demand at the lower price is $A - p_1$. Suppose that, for whatever reason, firm 1 is only willing to fill $x_1 < A - p_1$ worth of demand. Some of the unsatisfied demand may be willing to buy at price p_2 . Having been turned away at firm 1, do these consumers get a chance to go to firm 2? And, if so, how much demand turns up at firm 2? If we imagine, for example, that all consumers have identical utility functions and that firm 1 rations its output x_1 equally among all the consumers who show up, then it is "natural" to assume that the demand that moves to firm 2 is $A - p_2 - x_1$, as long as this is positive. (Why is this the natural solution? We won't go into details here, but this makes a good homework problem.) On the other hand, firm 1 might engage in the following rationing scheme: All the consumers who show up are given numbers. Then numbers are drawn randomly; the consumer whose number is drawn first gets to demand as much as he or she wants at p_1 . Then a second number is drawn, and so on, until all x_1 units are sold. This sort of rationing scheme leads to the conclusion that firm 2's demand is $(A - p_1 - x_1)(A - p_2)/(A - p_1)$. (Another good homework problem! In fact, here are two good homework problems, since it isn't so natural to assume that this is firm 2's demand if the good in question can be resold.) Moreover, when we analyze the conjectures of each firm about the other, we have to specify conjectures about how much demand each thinks the other will serve at various prices. This is too hard for now, so we will stick to the simple analysis given. But, if you have read this far, do take careful note: To make sense of this story, we had

to begin to be very specific about the “structure” of the marketplace. This, it turns out, will be the key to sorting through the various stories of classic oligopoly.

While we are here, it is a good time to mention another variation on the Bertrand story, which is also too hard for a first pass. This is the *Edgeworth-Bertrand* model, in which firms have Bertrand conjectures about each other (neither will change the price it charges), but each has capacity constraints on the amount of demand it can serve. Firms are assumed to fill all the demand that they can up to their capacity constraints. But there may be unsatisfied demand at some prices because of the capacity constraints. And then we need to specify what becomes of that demand. Does it go to the other firm, and in what quantity? Exactly the sorts of considerations given above intrude.

Another comment to make here is that the assumption that each firm is “forced” to serve all the demand that comes to its door is reasonable in the case we have been considering — constant marginal costs. This is so because no firm will (one expects) ever quote a price below its marginal cost, and so it is willing to serve any demand that does come along. But suppose we tried to redo our analysis in a setting where marginal costs were rising. To be very specific, suppose that we have two identical firms whose marginal costs of production at level x are x^2 . (That is, total cost of production of x units is $x^3/3$.) And suppose that the constant A in the demand function is 12. One Bertrand equilibrium, which is analogous to perfect competition, comes where price equals the marginal costs of the two firms. This is where $12 - (x_1 + x_2) = x_1^2 = x_2^2$, or (approximately) $x_1 = x_2 = 2.605$, for a price of 6.789. Note that at this Bertrand equilibrium each firm makes a positive profit, since it has strictly rising marginal costs, and price equals marginal cost at the equilibrium. Note also why this is a Bertrand equilibrium: If either firm raises its price, then it loses all its demand and makes zero profits. (It causes its rival to take substantial losses, but the firms in this story don't care about that.) If it lowers its price just a bit, it can (more than) double its demand. But every additional unit demanded is at a price below the marginal cost for that unit. So profits will go down.

Now consider the situation in which each firm quotes a price of 8, for total demand of four units, or two per firm. (This, it turns out, would be the industry supply if the two firms acted as a single monopoly firm, splitting production efficiently between them.) At this point, each firm has revenue of 16 and costs of $8/3$ for a profit of 13.333. Now if either firm raises its price from this point, it gets no demand at all. Whereas if a firm lowers its price a bit, it will (expect to) get a bit more than twice its demand. Ignoring the penny, if it shaves the price by a penny, it doubles its revenues to 32. But its costs shoot up to 21.333, for a net profit of 10.667. So we have another “Bertrand” equilibrium.

Of course, this second equilibrium supposed that a firm must meet all the demand that comes its way at the price it quotes. If the firm could shave its price by a penny from 8 and then only fill the orders it wishes to (up to where its marginal costs equal 8), then it would indeed increase its profits. So our second “Bertrand” equilibrium would collapse if firms could turn away demand. But then our first equilibrium would collapse as well. Recall that

there both firms quoted price 6.789. Neither firm will want to lower its price. But suppose that one of the two firms raised its price. The other firm, at a price of 6.789, would not wish to supply more than 2.605 units, since any additional units (at a price of 6.789) would lower its profits. (Marginal cost at 2.605 units is 6.789, recall.) So what demand would face the firm that raises its price? This depends, once again, on the rationing rule that the other firm uses to ration out the 2.605 units that it will sell. But whatever that rationing rule is, as long as the first firm sees some smooth fall in its output as it raises price, it will want to raise price to some extent. Why? Equate the first firm's marginal revenue to marginal cost and reexpress marginal revenue as price times $(1 + 1/\epsilon)$, where ϵ is the elasticity of demand facing the first firm as it raises its price. As long as the elasticity is finite (essentially, as long as its demand falls off smoothly) and marginal costs are nondecreasing, the first firm will raise its price from marginal cost. Most sensible rationing rules, including the two we discussed above, will have demand fall off smoothly. Thus price-equals-marginal-cost won't be a Bertrand equilibrium, when firms can refuse some demand and marginal costs are (strictly) increasing.

So what will be a Bertrand equilibrium? No pair of prices will do in this instance, at least if the firms have the same cost function. For suppose there were a pair of prices and a pair of quantities that would work. One case is that the two prices are the same. Then either that price is at the marginal cost of the two firms, and we saw that this wasn't possible, or it is above the marginal costs of the firms. This isn't possible either, since at least one of the two firms, by cutting price just a bit, can increase its demand discontinuously. As long as price is above marginal cost, profits will increase strictly by such a move. And if the two are quoting different prices, the lower price will have to be the monopoly price or above for one firm, since it acts “as if” it has the entire industry demand curve for itself.^b But then the firm with the higher price by cutting its price to just below that charged by the firm with the lower price will certainly do better. At this point, it is best that we leave Bertrand competition.

Kinked demand

The last classical model we will look at concerns kinked demand. A version of this is phrased in terms of “price conjectures,” and another is phrased in terms of “quantity conjectures.” We'll do the latter, as it makes sense a bit more easily.

In this model, the conjectures of each firm about the other depend on the equilibrium attained. Suppose we are in a situation where the two firms are producing, respectively, $\hat{x}_1$ and $\hat{x}_2$. At this point, firm 1 conjectures that firm 2 will continue to produce $\hat{x}_2$ as long as firm 1 produces $\hat{x}_2$ or less. But, firm 1 conjectures, if it attempts to produce an amount x_1 greater than $\hat{x}_2$, then firm 2 will increase its output to match

^b For rigor freaks, this implicitly assumes that the revenue curve in the industry is concave. Can you see the argument if this assumption doesn't hold?

whatever firm 1 produces. And firm 2 has symmetric conjectures about firm 1.

Where do *these* conjectures come from? They are, like all the other conjectures discussed so far, simply supposed. We've so far seen no reasons to support the hypothesis that they are reasonable conjectures. Which isn't to say that they aren't reasonable — each firm may well react in just this fashion. All we are saying is that we have no theoretical grounds to suppose that this will be so.

We note immediately that the consistency condition discussed back in the subsection on two von Stackelberg leaders plays a role here: The only points that are consistent with the firms' conjectures are where $\hat{x}_1 = \hat{x}_2$. This is so because, if $\hat{x}_1 < \hat{x}_2$ (say), then firm 2 conjectures that firm 1 will increase its quantity to $\hat{x}_2$ and symmetrically if $\hat{x}_2 < \hat{x}_1$. Accordingly, we can restrict attention to situations where the two firms pick identical levels of output, and we will use $\hat{x}$ to denote that level.

In this case, reaction functions can be used again. What will the reaction functions look like in this case? Suppose that firm 2 is at output level $\hat{x} = x_2$. Firm 1's problem, then, is to

$$\max_{x_1} (A - x_1 - y_2(x_1, \hat{x}) - k)x_1,$$

where $y_2(x_1, \hat{x}) = \max\{x_1, \hat{x}\}$. (That is, y_2 gives firm 2's supposed response to x_1 , which is to produce $x_2 = \hat{x}$ if firm 1 produces $\hat{x}$ or less and to match firm 1's output above $\hat{x}$.) This gives firm 1 a marginal revenue curve that is discontinuous. For $x_1 < x_2$, firm 1's marginal revenue is $A - x_2 - 2x_1$. For $x_1 > x_2$, firm 1's marginal revenue curve is $A - 4x_1$. Firm 1's optimal response is given by the level of x_1 at which marginal revenue equals marginal cost k . This is

$$x_1^*(x_2) = \begin{cases} (A - k - x_2)/2 & \text{if } (A - k - x_2)/2 \leq x_2 \text{ (i.e., } x_2 \geq (A - k)/3); \\ (A - k)/4 & \text{if } (A - k)/4 \geq x_2; \text{ and} \\ x_2 & \text{for } (A - k)/4 \leq x_2 \leq (A - k)/3. \end{cases}$$

We can draw some pictures of this muddle. First, for fixed $\hat{x} = x_2$, firm 1 sees its demand curve as the curve given in figure 10.4. Note well the kink — that is why this is called the theory of the kinked demand curve. The kink falls precisely at $x_1 = x_2$, which is where firm 2's (supposed) behavior changes. This demand curve gives the marginal revenue curve shown. Note the discontinuity in the marginal revenue curve, which falls at the position of the kink. And now we see why we get the funny reaction

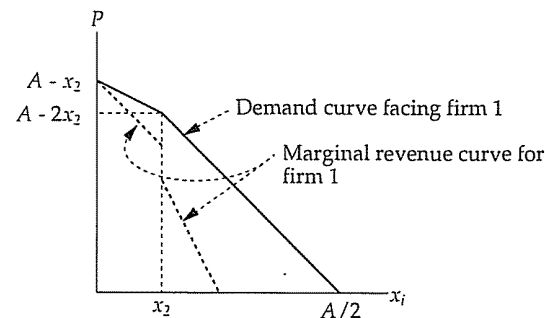


Figure 10.4. Demand and marginal revenue for the story of kinked demand.

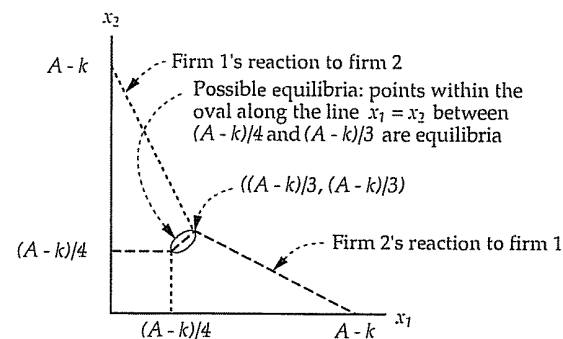


Figure 10.5. The two reaction curves for kinked demand.

curve described above (and drawn in figure 10.5). If x_2 is small enough, then the marginal revenue curve passes through the level k on its "bottom segment." If x_2 is rather large, then the marginal revenue curve passes through the level of k on its "top segment." And for a range of x_2 , marginal revenue jumps from above k to below k at its discontinuity point. For those values of x_2 , the optimal response of the firm 1 is just to match firm 2's output.

So what are the equilibria? We have also drawn on figure 10.5 the reaction function of firm 2 to firm 1's choice of quantity output. The two intersect at a number of points, namely $x_1 = x_2 (= \hat{x})$ between $(A - k)/4$ and $(A - k)/3$. Any of these give an equilibrium. Note that this ranges from the monopoly outcome (total supply is $(A - k)/2$ and price is $(A + k)/2$) down to the Cournot outcome (total supply is $2(A - k)/3$ and price is $A/3 + 2k/3$).

Which of the classic models of duopoly is correct?

We have seen that the models of Cournot, Bertrand, and von Stackelberg all make distinctly different predictions as to what will happen in the duopoly situation we describe. The model of kinked demand makes a number of different predictions, any of which might happen, which widens the set of predictions still further. Which, if any, of these predictions will be borne out by data?

In the end, the answer to this question must be empirical. And it will probably be that none of the models is correct in all cases of real oligopoly. Instead, one may be able to develop propositions such as: Across a set of industries, long-run equilibria more closely resemble the predictions of Cournot, say, the more the industry has the characteristics _____ (fill in the blank). To form a better test of such propositions, one would like to see why, on theoretical grounds, certain characteristics might go well with certain of the classic models.

The sorts of characteristics that might be studied are those concerned with the actual mechanics by which price is set and demand is filled in an industry. The classic models above are quite deficient in this respect. Demand is given, and equilibrium conjectures about out-of-equilibrium responses are specified. But (except for when we discussed rationing in the Bertrand model) no attention is paid to how equilibria are achieved.

To take five extreme cases, suppose that the mechanics of the industry are one of the following:

(a) Two producers work in isolation, preparing the quantity they bring to market. They decide upon these quantities with a foreknowledge of this market structure and with knowledge of the characteristics of demand, but neither side gets to see how much the other is producing. After production is completed, each side brings the quantity it produced to a central market place, where it is handed over to a "state sales commission" that sets price so that the total amount brought to market is just demanded at the set price.

(b) We have the same two producers as above and the same state sales commission, but one producer is able to produce its quantity first. That is, this producer brings to the market a quantity x_1 , which the second producer observes. And then the second producer decides on an amount x_2 to produce, knowing that price will be set so that the market clears at total quantity $x_1 + x_2$.

(c) Things work much differently in this case. The two producers independently call the state sales commission and give a price at which each

is willing to sell. The state sales commission investigates market demand at those prices, and then calls back the two producers, telling them how much they are obligated to bring to the market. (Or, to take a variation, the state sales commission tells the one quoting a lower price how much it may bring, if it wishes. After the first says how much demand it will service, the state sales commission calls the higher price producer. The state sales commission has a particular rationing scheme it uses in cases where a firm doesn't wish to service all the demand it gets.)

(d) Back to our two producers who produce quantities independently. But now they call the state marketing commission in advance, saying how much they plan to bring. And if one says it will bring more than the other, the second is given permission by the state marketing commission to bring up to the amount being brought by the first.

(e) This one is very complex. The two producers must first install capacity for production. Capacity is costly, coming at a cost of k_1 per unit capacity installed. Capacity is installed by the two simultaneously and independently. Then each sees how much capacity the other installed. And then each calls the state marketing commission and quotes a price at which it is willing to sell. Production is possible up to the level of capacity at an additional marginal cost of k_2 per unit where $k = k_1 + k_2$. No production beyond the level of capacity is possible. If a capacity constraint does bind when prices are named, the state marketing commission rations demand by some system or other. This good is resaleable among consumers, and all consumers are aware of market demand conditions.

In scenario (a), Cournot conjectures make particularly good sense. Each side believes that the other won't change its quantity in the face of a "change" by the one for the simple reason that neither side has the opportunity to change its quantity. In scenario (b), von Stackelberg seems sensible. The second firm (presumably) takes the first firm's quantity supply as given, since it is given, and maximizes its own profits. So it works off its reaction curve. And the first firm, if it understands this, would pick an initial quantity so that it maximizes its profits with the second firm's optimal response coming from that reaction curve. Scenario (c) leads naturally to Bertrand conjectures for the same reason that scenario (a) leads to Cournot. Scenario (d) may remind you of kinked demand, but it isn't quite the same; we have to check that if one firm outproduces the second the second will wish to respond by increasing its output unit for unit. (Any guesses about how this will turn out? Sounds like a good homework problem.) And scenario (e) probably reminds you of nothing you've ever

seen before. But, it turns out, it leads fairly naturally to a prediction of Cournot outcomes. Seeing that this is so is fairly difficult.^c

Do any of these models match the market mechanics of any industry you can think of? Probably not. For one (important) thing, each of them is a story of a one-shot encounter between two firms. Any of them could be complicated by adding, "... and the same thing happens again every month, with the two firms discounting profits at a rate of (say) 10% per year." We will see in chapter 14 that this complication is extraordinarily significant.

The answer to the question posed at the start of this subsection is that no one of the classic models is going to give accurate predictions in every case. Each makes sense (and should give accurate predictions, although this remains an empirical question) if you have in mind a particular detailed mechanism for how the market operates. None will make sense if you have in mind some different mechanisms. You *might* be able to tell whether (or when) one model or another should give a good prediction by looking at the details of the market mechanism. But this will require a theory of how to move from market mechanics to outcomes.

We will use techniques from noncooperative game theory for this purpose. The key to these techniques is that one is very precise about the mechanics of the competitive interaction, saying precisely who moves when, with how much information. By being precise in this fashion, one *sometimes* gets quite sharp predictions. (As we'll see, at other times the predictions are anything but sharp.) We will begin to develop these techniques in chapter 11, and we will return to the classic models of duopoly at the end of chapter 12. But before doing so, we discuss a few further basic ideas from the theory of imperfect competition.

10.2. Bibliographic notes and discussion

The treatment of duopoly with undifferentiated commodities that we have given is adequate for our present purpose of developing the basic tools of microeconomic theory. But this treatment is both a bit unfair to the classical references themselves, which give a much more reasoned explanation of from where the sundry conjectures might come and more than a bit incomplete as a sampling of what is important in the theory of oligopoly and, more generally, imperfect competition. The reader can repair the injustice by reading the classics: Cournot (1838), Bertrand (1883),

^c And if after the next two chapters you wish to see how it works, see Kreps and Scheinkman (1983). Good luck.

von Stackelberg (1934), Hall and Hitch (1939) and Sweezy (1939) (the last two for kinked demand). And, preferably after completing this book, the reader can plunge into the rich theory of imperfect competition quite efficiently by studying Tirole (1988). (By recommending Tirole alone, I do not mean to slight other books. But Tirole gives a comprehensive treatment of most important theoretical issues, with everything done in the most modern style, a style for which the rest of this book is meant to prepare you.)

We will in later chapters return to some of the issues connected with oligopoly and imperfect competition. But these are issues that take many pages to do justice to, so for the most part we do not pursue important variations on the basic models above. To whet your appetite, then, we close with the mention of some of the more important variations.

Oligopoly

We examined only the case of duopoly and, at that, the case of two identical firms. Many oligopolies have more than two firms and/or firms that are not identical; it is obviously germane to extend our models accordingly. Solving for the N -firm symmetric Cournot and Bertrand equilibria, where the firms are identical, demand curves are linear, and production technologies are linear, is an entirely straightforward exercise. (In fact, it is problem 5.) Looking for asymmetric equilibria, which is necessary when firms are not identical, is a bit, but only a bit, harder (cf. problems 6 and 7).

Entry and exit

Why are there only a given number of firms in an oligopoly? Especially when firms in the industry are making profits, we can wonder why the industry doesn't attract entry. In the case of monopoly, one can make a case based on legal restrictions. For oligopolies, similar cases can be made in some instances — only a certain number of firms may be licensed to make a particular product. These licenses may come from the government, say in the production of pharmaceuticals or pesticides or in VHF broadcasting, or they may be granted by a monopoly holder of a patented process.³

But more often the explanation is that there are economic (instead of legal) *barriers to entry*. The threat of cut-throat competition by incumbent

³ Why would a patent holder ever license another firm to produce the product on which it holds the patent, instead of producing the product as a monopolist? For one story, see problem 19 of chapter 12.

firms can hold other firms out of the industry. (When spun out into a model, this story becomes a more sophisticated version of story of entry deterrence in monopoly that was caricatured in chapter 9.) There may be technological or "knowledge" barriers to entry; breaking into the business of manufacturing large commercial jet airframes is, to say the least, difficult for a firm that has no experience making jet aircraft. And fixed costs by themselves may prove to be barriers to entry — see problem 8 below.

Taking a more detailed look at an industry, there may be segments, some of which are inhabited by some firms, others inhabited by others. Then the notion of an entry barrier is transformed into the notion of a mobility barrier — something that prevents firm A from moving into the profitable sector of the industry inhabited by firm B. For a normative discussion of entry and mobility barriers, see Porter (1980).⁴

When asking why there are only four or five (or however many) firms in a particular industry, one can think of the industry as static; the focus is on a long-term stationary situation, in which no new firms enter (and no incumbent firms leave). In a sense, these are not theories of entry, but theories of no-entry. A different perspective comes from looking at an industry or a product more dynamically; think of industries that grow and industries that decline. There have been a number of studies (of varying degrees of formality) about entry and exit decisions of firms in such industries, studies of which firms enter at which stage of industry growth, and which depart as the market shrinks away.

Differentiated commodities

In the formal models discussed in this chapter, the duopolists were assumed to be producing and selling undifferentiated commodities. Note especially that it was the undifferentiated character of the product that led to the extreme results of Bertrand competition, since a penny less in price by one firm meant that firm got all the demand.

Some oligopolies produce virtually undifferentiated commodities — producers of industrial chemicals, for example.⁵ But in many oligopolies, the products sold are (or are made to seem) different. A Chevrolet and a Ford or a Toyota and a Honda of roughly the same size and character may be similar, but consumers perceive differences between them; if

⁴ That is, Porter is writing for oligopolists, explaining how to build and maintain entry and mobility barriers.

⁵ Even in this case there is some differentiation in the "goods" having to do with relationships between the producer and the client firm. The test of how much differentiation exists is quite simple: How much lower a price would one firm have to charge than all others before it stole virtually the entire market from its rivals?

Toyota charges \$500 less than Honda for a similar car, it may increase its demand at Honda's expense, but some loyal customers will still stick with Honda.

If we just take as given that oligopolists produce differentiated commodities where the differences (and corresponding cross-elasticities of demand) are exogenously imposed, we can redo much of the simple duopoly theory we developed above. (You will do this in problem 9.) But it is even more interesting to consider whether oligopolists will endogenously strive to differentiate their products. The reason to engage in differentiation should be clear: Especially if we believe in the Bertrand model, but in any model where oligopolists "compete," the more they differentiate their goods from one another, the less competition there will be, and the more each can engage in something like monopoly pricing. (See problem 10 for one example.) On the other hand, if your rival has fixed his product in a certain "position" in terms of its attributes, by moving the characteristics of your product closer to his you are providing for yourself a broader market. Starting with Hotelling (1929), there has been great interest in the extent to which oligopolists engage in willful product differentiation and whether the amount of product variety provided is more or less than is socially desirable. Tirole (1988, chap. 7) gives a good survey.

Research and development

Nowhere in this book have we (or, except at the very end, will we) think about the dynamic process of research and development in which new products and production technologies are discovered, developed, and refined. These issues transcend the setting of oligopoly. We can investigate how much R&D will be undertaken in competitive industries and by monopolies, but oligopoly is a particularly fertile field for inquiry into these questions, since the rivalry between oligopolists is meant to provide strong incentives to engage in R&D (incentives that are weaker at best in monopoly, except for the cases where patents reward successful R&D with further monopoly). At the same time, oligopolists have the wherewithal to engage in large scale R&D. As usual, our suggestion is to begin to study these topics with Tirole (1988, chap. 10).

Monopolistic competition

We come in the end to the "other" model of competition between imperfectly competitive firms, the model of *monopolistic competition* as developed by Chamberlin (1933). Most books on intermediate microeconomics (and many principles books) develop this theory in detail, as much detail or more than is given to oligopoly. In comparison with oligopoly, the theory is remarkably

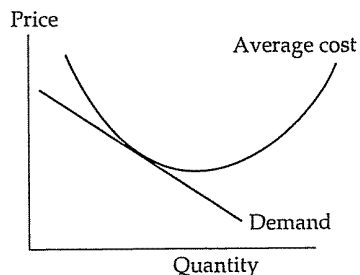


Figure 10.6. Equilibrium in a monopolistically competitive industry.

determinate — it gives simple and clean answers complete with a very interesting picture (figure 10.6). Were it not for the presence of this theory in most lower level texts we would ignore it here altogether. But since many readers will have seen it and wonder how it fits into the general picture being presented here, we will offer a very fast recapitulation and then some criticisms.

The story of monopolistic competition runs as follows: We imagine an industry for a product that can be produced in many different varieties. There are many firms in the industry and many more waiting on the sidelines to enter if entry seems profitable. Each firm produces a particular (unique) variety of the product in question. Each firm's technology is described by a U-shaped average cost curve, given, say, by a fixed cost plus increasing marginal cost technology. If we compute demand curves for each firm by varying the price charged by the firm while holding other prices fixed, each firm faces a downward sloping demand curve. This downward sloping demand curve arises because each firm produces a product that has unique characteristics and so attracts (to some extent) its own clientele.

Each firm believes that it can adjust its price/quantity to be any point along its own demand curve without engendering a change in price for any of its many rivals, so each approaches its own demand curve as would a monopolist; each sets marginal revenue equal to marginal cost, finding the price to set and quantity to sell. The usual justification for this assumption is that if one firm lowers its price all other firms see a very small change in their demand and so will not react at all.

As for entry and exit, if any firm in this industry is making losses, it will leave the industry. And if any firm in this industry is making positive profits, entry to the industry will occur. These entrants all pick their own unique variety of the good to sell, and the entry of any one entrant has negligible effect on the demand curve of any firm already in the industry. But as many entrants enter, the effect is cumulative on industry participants; they see their demand curves shift down and to the left.

Taking this story at face value, what will an equilibrium look like? Every active firm in this industry will be making zero profits; that must be true because firms making negative profits leave, and if firms are making positive

profits, entry occurs. If an equilibrium is a situation with no inflow or outflow of firms, the only possibility left is zero profits. This means that the demand curve each firm faces must be tangent to and otherwise below the firm's average cost curve; if the demand curve lies entirely below the average cost curve, the firm will be making losses no matter what it does, whereas if the demand curve is (at any level of quantity) above average costs, the firm can make strictly positive profits at that quantity. If in equilibrium firms face downward sloping demand, the only possible picture is that given in figure 10.6. The firm's demand curve is tangent to average cost at a quantity that is less than efficient scale. To speak in jargon, firms are producing (in equilibrium) at a level at which they have *excess capacity*.

Following development of the rudiments of this theory, it is normal to comment on the "efficiency" of this market structure; on the one hand, firms are producing at an inefficient scale (bad), but consumers are getting something for this, namely a wide variety of products (good). The issue of efficiency of a monopolistically competitive industry gets rather complex, and we won't attempt to summarize the debate here. (See Tirole, 1988, chap. 7.) Instead we ask, What industry might this be a model of?

The classic answer given is retailing or service industries. The feature that is meant to differentiate the many shops is in the first place location; each shop is unique in that it alone occupies a particular location in a particular block, and it faces a downward sloping demand curve despite the vast number of shops in the world, because it "really" only competes with shops near to it. Someone living on Second Avenue between 10th and 11th Streets is unlikely to consider a visit to a shop all the way across town, even if that shop across town has the product at a much cheaper price than any of the local shops. The real choice set for this consumer consists of the shops on the corners of Second Avenue and 10th and Second and 11th, and maybe, if it really has good prices, the shop on Third and 10th. Hence each shop, even if it raises its price a bit above that of the competition, will still attract some clientele.

Whether the retailing business approximates the conclusions of monopolistic competition is an empirical question. Retailing in the United States, at least, is usually a very low margin business, which is a point in favor of the theory's implications. But it is worth objecting to the application of the model to shops on various street corners on purely theoretical grounds. (That is, we may have the right conclusions but for the wrong theoretical reasons.) The objection is that what the shop on Second and 10th does cannot be supposed to have negligible effect on how nearby shops fare and what they do. The shop on Second and 11th knows well that its competition is just a block away, and if Second and 10th runs a special on apples, Second and 11th will respond with a special on bananas. Moreover, if the shops on Second Avenue are making profits and a new shop, in response to this, sets up on Second Avenue and 12th, this will significantly affect the demand of the Second Avenue shops (that are near to 12th Street), and it will have little or no effect on the demand at shops across town. This situation is not as described in the model of monopolistic competition; it is a model of local oligopoly with entry.

There is nothing wrong with having a theory of local oligopoly, except that just as oligopoly theory isn't determinant, neither is a theory of local

oligopoly. In the story of monopolistic competition, each shop owner believes he can change his prices without response from any competitor. The reason given is that when he changes his prices, none of his competitors see much effect on his own demand. That clearly is wrong for local oligopolists. We could assume in a local oligopoly that all the shop owners have Bertrand conjectures, but (at this point) that is just one assumption out of many that we could make.

Is there an industry that meets the conditions of monopolistic competition? We leave the reader with this question as a challenge. Can you think of an industry in which (a) firms face downward sloping demand, (b) cross-elasticities of demand for any pair of firms are low, and (c) there is free entry? Here is one case to chew on: Consider the lunchtime restaurant trade (or private clubs) in a metropolitan business district. Can you construct a model of consumer demand at a large number of restaurants such that if any restaurant lowers its price it draws a little bit of clientele from each of the other restaurants, and such that a new restaurant entering the business takes a bit away from all incumbents, but doesn't drive any one (or a few) significantly closer to bankruptcy?^d

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^d I am grateful to Drew Fudenberg for suggesting this industry as a possibility. If you wish some help in constructing preferences with the required properties and a bit of heavy mathematics won't dissuade you, see Hart (1985), from whom much of this commentary is drawn.

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10.3. Problems

- 1. In the model of a duopoly with undifferentiated commodities, one firm has Cournot conjectures and the second has Bertrand conjectures. Can you find an equilibrium?
- 2. In the model of kinked demand, the firms' conjectures were such that the only possible equilibria were symmetric. If we change the nature of the firms' conjectures, then perhaps we can find asymmetric equilibria. For both sorts of conjectures described below, find all the pairs $(\hat{x}_1, \hat{x}_2)$ that are equilibria.
 - (a) At equilibrium $(\hat{x}_1, \hat{x}_2)$, firm 1 believes that firm 2 will continue to produce $\hat{x}_2$ as long as $x_1 < \hat{x}_1$, but that firm 2 will match any increase in firm 1's output (above $\hat{x}_1$) unit for unit, and vice versa. That is, firm 1 believes that firm 2 will produce $\hat{x}_2 + \max\{0, x_1 - \hat{x}_1\}$. Firm 2 holds symmetric conjectures about firm 1.
 - (b) At equilibrium $(\hat{x}_1, \hat{x}_2)$, firm 1 believes that firm 2 will continue to produce $\hat{x}_2$ as long as firm one produces a level x_1 below $\hat{x}_1$. But firm 2 will increase its production in response to an increase in firm 1's production in an amount that keeps the relative market shares of the two firms the same as in equilibrium. That is, firm 1 believes that firm 2 will produce $\max\{\hat{x}_2, (\hat{x}_2/\hat{x}_1)x_1\}$. Firm 2 holds symmetric conjectures about firm 1.
- 3. An economy I know has 1,000 identical consumers and two goods. The two goods are phuffle, a nonnarcotic stimulant, and manna, a basic foodstuff. Each consumer has a utility for phuffle and manna of the form $u(x, z) = w(x) + z$, where x is the amount of phuffle consumed and z the amount of manna. Assume that w is strictly concave and continuously differentiable. The function w is concave and satisfies $w'(0) = 100$ and $w'(100) = 1$. The price of manna is always \$1, and each consumer has wealth exceeding \$10,000 which is spent entirely on phuffle and manna.

(a) What is the shape of the market demand curve for phiffle, for prices between \$1 and \$100?

Two firms sell phiffle. For some reason, one firm is willing to sell at a price of \$2, while the other insists on a price of \$3. Moreover, the first firm is willing to sell only 50,000 units of phiffle. The second firm is willing to sell to all comers. Assume that $w'(90) = 2$ and $w'(75) = 3$.

(b) Suppose that the following mechanism is used for the distribution of phiffle. The first firm asks each consumer how much phiffle he wishes to buy at a price of \$2. If total orders are less than 50,000, each consumer is given whatever he asks for. Otherwise, phiffle from firm 1 is rationed equally to each person who wants some, up to the amount that the person wants. That is, we divide the 50,000 units equally among all the consumers and ask whether any consumer is getting more than he asked for. If so, those consumers who asked for less are given what they asked for, and the remainder of their share is divided equally among all the consumers who are still rationed, and so on.) After this rationing scheme, consumers can go to the second firm, if they wish, and buy as much (more) phiffle as they would like for \$3. What will be the result of this scheme?

(c) Suppose that phiffle is distributed as follows. The first firm asks each consumer how much phiffle she wishes and then distributes the phiffle as follows: A consumer is chosen at random and is given whatever she asks for, up to 50,000 units. Then, if anything remains of the 50,000 units, a second consumer is chosen at random, and so on, until either all consumers have been served or the 50,000 units are exhausted. After this distribution, consumers can purchase phiffle from the other firm, at \$3 per unit. What will be the outcome of this distribution system if phiffle cannot be resold?

(d) What will be the outcome of the distribution system in part (c) if phiffle can be resold? (Assume that consumers in this economy are very savvy folks.)

(e) If you like a challenge, think about these three scenarios with the added complication that the economy is populated with a large number of heterogeneous consumers. You will need to formulate some of the problem for yourself.

■ 4. Go back to a manufacturing monopoly that produces items at a constant unit cost k , and that must sell these units to retailers who sell the items at zero additional cost (beyond the cost the retailers incur to buy the items from the monopoly). Suppose there are two retailers, and they act according to Cournot conjectures. Demand is (what else) $P = A - Q$.

(a) Assume the monopoly charges retailers according to simple linear prices — at a price the monopoly sets. What is the optimal price for the monopoly to set?

(b) Suppose the monopoly can use a two-part tariff scheme, where it charges each retailer a fixed fee, plus a per unit charge for any units bought. What is the optimal pricing scheme for the monopoly?

(c) Now consider what will happen if the two duopolists compete with Bertrand conjectures.

■ 5. Consider an industry in which N firms all produce an undifferentiated product. Demand for the product is given by $X = A - P$. Each producer is identical, having a constant average cost of k .

(a) Suppose all N producers have Cournot conjectures: Each conjectures that it can change the amount it produces and all its rivals will continue to produce the amounts they are producing. What is the symmetric equilibrium in this case? (An equilibrium is symmetric if all the firms are producing the same amount.) What happens to the equilibrium price as N approaches infinity?

(b) Suppose all N producers have Bertrand conjectures: Each conjectures that it can change the price it charges without any reaction from rivals in the prices they charge. What is the symmetric equilibrium in this case?

(c) Suppose one firm is a Stackelberg leader and all the rest have Cournot conjectures. That is, each of the $N - 1$ followers believes that it can change its quantity without any response in quantities from any of the other firms. The one leader understands this and picks its quantity optimally. What equilibria can you find? What happens to price as N goes to infinity?

(d) Suppose the N firms are numbered $1, 2, \dots, N$, and they have the following sort of conjectures: Firm N has Cournot conjectures. Firm $N - 1$ has Cournot conjectures about firms 1 through $N - 2$ and believes that firm N works off its Cournot reaction curve. Firm n has Cournot conjectures about firms 1 through $n - 1$ and believes that firms $n + 1$ through N work off the reaction curves given by their conjectures. What equilibria can you find? What happens to price as N goes to infinity?

(e) Construct a model in the spirit of kinked demand for the setting of an N firm oligopoly.

■ 6. Here is an alternative scheme for finding Cournot equilibria. In our discussion in the chapter, we computed reaction functions for each firm

of the form "what quantity x would the firm choose to output if the other firm produced x' ?" Suppose instead we answer the question: What quantity x produced by one firm is consistent with X in total from both firms? We would find this by looking at the problem

$$\max_{\delta} (A - k - X - \delta)(x + \delta)$$

and finding, for a given X that amount x that has $\delta = 0$ as a solution. Why? Because we are computing here how the firm would choose to vary its output from x if the total output is X when the firm chooses x ; when a variation of 0 is optimal, x is consistent with X . Carrying this out (so you will see that it works), the first-order conditions in δ are $A - k - X - x = 2\delta$, which are satisfied at $\delta = 0$ if $A - k - X = x$. That is, $x(X) = A - k - X$ is the production by one firm consistent with X produced by both. At an equilibrium $2x(X) = X$, or $2(A - k - X) = X$, or $2(A - k) = 3X$, or $X = 2(A - k)/3$, which then yields $x = (A - k)/3$, our Cournot equilibrium. (This technique is due to Novshek [1984].)

(a) In part (a) of problem 5, you were asked to find the symmetric equilibrium of an N firm oligopoly where firms have Cournot conjectures. Using the technique just presented, find all the equilibria including any asymmetric ones that may exist.

(b) Now repeat part (a) of this problem but in a setting where the N firms that make up the oligopoly have possibly different unit costs; let k_n be the unit cost of firm n for each n .

■ 7. Suppose that in a duopoly with an undifferentiated product, demand is given by $X = A - P$ and firms have Bertrand conjectures and different unit costs. Specifically, firm 1 has constant average costs of k_1 and firm 2 has constant average costs of k_2 , with $k_1 < k_2$. What is (are?) the equilibrium(a)?

■ 8. Suppose an industry produces an undifferentiated product for which market demand is given by $X = A - P$. There are many potential producers for this product, each of whom has a production function of the form: Fixed costs of F must be paid for being in business, and the marginal cost of a unit of production is a constant k . We imagine that firms decide whether to enter the industry under the supposition that, after all the firms that are going to enter do so, competition will be according to the Cournot model. That is, if N firms are in the market each has Cournot conjectures. An equilibrium is achieved with N firms in the industry if

each firm, having its Cournot conjectures, does no worse than break even, whereas if another firm entered and made this an $N + 1$ firm Cournot oligopoly, all the firms would lose money. What is the equilibrium in this case? What (if anything) would be the equilibrium if firms had Bertrand conjectures throughout? (Note well: The number of firms is set and then firms compete. The exercise is not that firms charge a price, say, and then other firms can enter assuming firms already in the industry will stick to that price. But as long as we're sketching this alternative, what would be the Cournot and Bertrand equilibria with entry under this alternative scenario?)

■ 9. Suppose two duopolists produce given differentiated goods. The prices that each obtains in equilibrium depends on the quantities output by each (or, alternatively, the equilibrium quantities demanded for each depends on the prices each charges), according to the inverse demand functions

$$p_1 = A - x_1 - bx_2 \quad \text{and} \quad p_2 = A - x_2 - bx_1,$$

for constants A and b . The constant b is restricted to the interval $(-1, 1)$, where $b > 0$ means that the goods in question are "substitutes" and $b < 0$ means they are "complements." (We use quote marks here because we are not working with precise definitions of these terms; cf. chapter 2.)

(a) Give Cournot, von Stackelberg and Bertrand equilibria for this model. If you find the algebra getting a bit too thick, solve for equilibrium values (quantities, prices, profits) for the parameters $A = 10$, $k = 1$ and $b = .9, .5, .1$, and $-.5$. Note that the two firms do better in Bertrand than in Cournot for the case $b < 0$. Can you give an intuitive explanation of this?

(b) In the von Stackelberg equilibrium of part (a), the follower has Cournot conjectures and the leader optimizes given the follower's reaction function. We could equally well imagine a case in which the follower had Bertrand conjectures and the leader optimizes given the reaction function so entailed. This is called the *Bertrand-von Stackelberg* equilibrium. (The von Stackelberg equilibrium in part (a) is then called the *Cournot-von Stackelberg* equilibrium.) Solve for it, at least for the four parameterizations given previously.

(c) In both sorts of von Stackelberg equilibria, the leader has higher profits than if there were no leader at all. Prove that this is so. (It will suffice to demonstrate a weak inequality.)

(d) In Cournot-von Stackelberg for $b > 0$ and Bertrand-von Stackelberg for $b < 0$, the follower has smaller profits than if there is no leader, while for Bertrand for $b > 0$ and Cournot for $b < 0$, the follower has larger profits than if there is no leader. What intuitive explanations can you give for these observations?

This entire problem is the tip of an iceberg that goes by the rubric "strategic substitutes and complements." If you read Tirole (1988) or other treatments of imperfect competition, watch for it under that name.

■ 10. Imagine a duopoly in which firms produce an "identical good" for consumers who live along a 100-mile highway. We imagine (in the spirit of problem 10 in chapter 9) that consumers are distributed uniformly along the highway at a density of ten consumers per mile. Each consumer has reservation price \$100 for one unit of the good, but each pays a cost getting to and from the store that must be added to the price of the good when comparing against the reservation price. The good costs each of the two firms \$1 per unit to produce. Each firm is allowed to open one store somewhere along the highway. If the firm is located d miles from a consumer, the consumer pays $\$(d/50)^2$ for the round-trip from home to the store and back. The two firms locate their stores (which serves to differentiate their goods in the minds of consumers, if they don't locate one on top of the other) and then they compete with Bertrand conjectures. What is the equilibrium in this case? (We have been a bit imprecise in specifying the model, so be very clear what assumptions you are making. If you like a challenge, consider what would happen if travel costs were linear instead of quadratic. If you do this, be prepared for enormous frustrations.)

Further problems on variations on the classical models of duopoly are given in chapter 12.

part III

Noncooperative game theory

chapter eleven

Modeling competitive situations

Prologue to part III

In part III, you will be introduced to some of the terms and concepts of noncooperative game theory. Note the words “introduce” and “some”; this is far from rigorous and far from comprehensive. But I hope it will serve as enough background for the sorts of applications we will discuss later and will arouse your curiosity sufficiently so that you’ll study it further.

Note also at the outset that these chapters will concentrate exclusively on *noncooperative* game theory, and, at that, primarily on equilibrium (or Nash equilibrium) concepts. A lot of game theory, perhaps the bulk of developments from the 1950s and 1960s, concerns so-called cooperative game theory. Recently, however, the emphasis has shifted toward noncooperative game theory — to the point where noncooperative game theory has become a very important tool for microeconomists. Do not be misled by the terminology. It isn’t that economists have recently become more interested in uncooperative behavior. Rather, as we will see in great detail in chapter 14, the emphasis recently has been more on explaining (how) cooperation (and many other forms of aggregate behavior) can (emerge) from (self-interested) individual (behavior within a given set of “rules,” which is the defining characteristic of noncooperative game theory. It is probably best just to regard the terminology as jargon and wait to see what emerges from so-called noncooperative theory. Nevertheless, the jargon is used, so be forewarned that we will not discuss a large chunk of game theory at all. Seek other sources for an introduction to cooperative game theory.

In this chapter, we will look at how competitive situations are conventionally modeled in noncooperative game theory. Two sorts or *forms* of models are used, the so-called *extensive form* game and the *normal* or *strategic* form game. We will describe each of these modeling forms in this chapter and consider some of the basic issues involved in moving from

one form to the other. In chapter 12, we will consider solution concepts for the games posed in this chapter, concentrating on *Nash equilibrium* analysis. Then in chapters 13, 14, and 15 we investigate three important types of application of the theory: models of *incomplete information*, *repeated play*, and *bilateral bargaining*.

11.1. Games in extensive form: An example

The story

Consider the following story. As owner-manager of Jokx Toys and Games, you are thinking about introducing a new game called Oligopoly, which will teach children from ages 8 to 12 the basic principles of imperfect competition. You must decide very soon whether to introduce Oligopoly and if you do decide to go ahead, you will have to spend \$40,000 to complete design of the game, advertise it, and set up production.

The market for a game like Oligopoly is highly uncertain. You decide to consider two possibilities: The market will either be large, yielding total sales of 20,000 units, or it will be small, with sales totalling 6,000 units. You assess probabilities .4 and .6 for these two possibilities. These figures suppose a wholesale price per unit of \$12; raising the price will cause sales to plummet, whereas lowering it will not increase demand appreciably.

Another source of uncertainty is that a competitive firm, Beljeau Games and Toys, is considering the introduction of a game called Reaganomics, which will compete directly with Oligopoly. In fact, if you introduce Oligopoly and Beljeau introduces Reaganomics, the competition that ensues will force each of you to charge only \$10 per unit (wholesale) — the overall market, which will be either 20,000 or 6,000 units, will not be enlarged by the fall in price — and you will each get a one-half share of the market.

It will cost you \$5 per unit to produce Oligopoly in addition to the \$40,000 fixed costs mentioned above. You will be able to produce exactly as many units as you sell (produce to demand).

The situation for Beljeau is somewhat similar. As you sit debating whether to introduce Oligopoly, the managers at Beljeau are debating whether to introduce Reaganomics. The press of getting these games out in time for Christmas means that you cannot wait to see what they do before deciding whether to go ahead with Oligopoly; similarly, they must decide about Reaganomics before learning of your decision on Oligopoly. But they have one advantage — they have commissioned a market survey whose results they will learn before deciding whether to proceed with

11.1. Games in extensive form: An example

Reaganomics, a survey that will tell them without error whether the market will be large or small.

Beljeau will incur a fixed cost of \$60,000 if it develops Reaganomics, and a unit cost of \$3 per unit (produced to demand). Reaganomics will sell for precisely the same price as Oligopoly — \$12 wholesale if only one product is in the market and \$10 wholesale if both firms are marketing their (respective) games.

Decision trees

Looking at this problem from your point of view, we can build a *decision tree* that represents the problem you face. The tree is given in figure 11.1. For those of you who have never seen a decision tree before, we offer some words of translation. We start at the left-hand side; the box there with two branches coming out represents the decision that you must make right now whether or not to introduce Oligopoly. Boxes are called *decision* or *choice nodes*. If you choose to introduce Oligopoly, then things will begin to happen to you: You learn the size of the market (large or small), and you learn what Beljeau decided to do (introduce Reaganomics or not). These are things outside your control; from your point of view they are “chance events” (although you may have a pretty good idea what Beljeau will do). So we depict them as *chance nodes*, or branching points in the tree marked by circles. On the other hand, if you don't introduce Oligopoly, from your point of view the problem is over.

This gives us five “branches” in the tree, each one representing a unique sequence of choices by you (Jokx) and outcomes of events that are outside your control. Note in this regard that we put in two chance nodes for Beljeau's choice, so that we have four outcomes following a decision by you to enter the market. Then for each of the five branches, we can

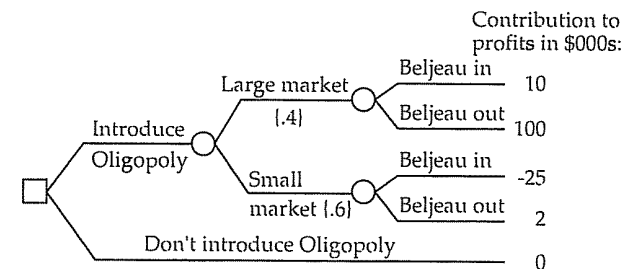


Figure 11.1. The decision tree of Jokx.

evaluate how much contribution to profit you will receive. For example, if you introduce Oligopoly, the market is large, and Beljeau introduces Reaganomics (the topmost branch), you will sell 10,000 units for \$10 wholesale, or \$100,000 in revenues, less fixed costs of \$40,000 and manufacturing costs of $(10,000)(\$5)$, or total costs of \$90,000, for a \$10,000 contribution to profits, and so on. These numbers are written in on figure 11.1 at the end of each branch. Finally, we know the probabilities that the market is large or small, so we put these in on the appropriate branches in the tree.

Could we depict your decision tree in other ways? Figure 11.2 gives two possibilities. In figure 11.2(a) we interchange the two chance nodes, putting Beljeau's decision first and then the size of the market. And in figure 11.2(b) we combine the two chance nodes into one having four possible outcomes. The rule that we must follow in constructing decisions trees is

The fundamental rule of decision trees. A chance node precedes a choice node in the tree if and only if the uncertainty represented by that chance node resolves

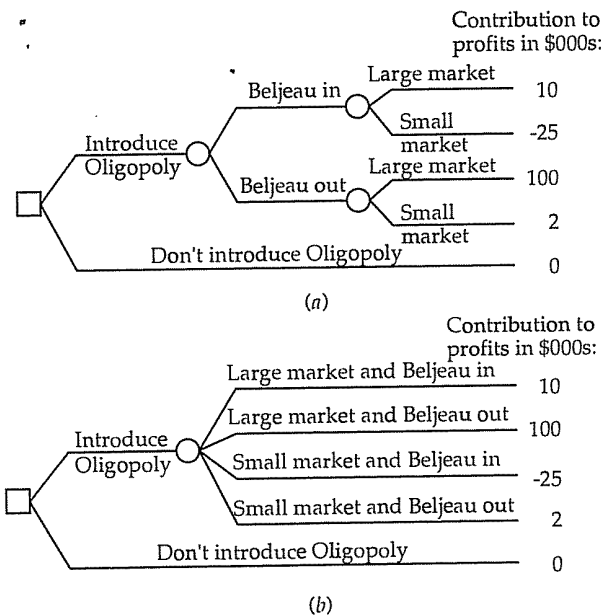


Figure 11.2. Two different representations of Jokx' tree.

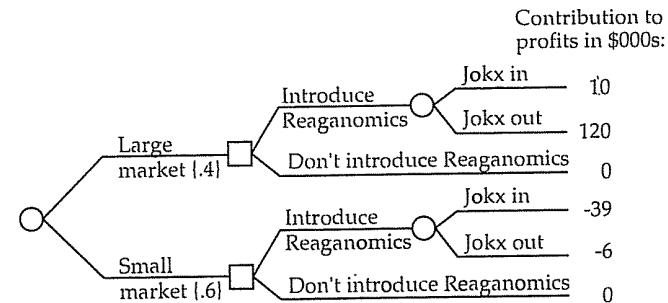


Figure 11.3 The decision tree of Beljeau.

in the mind of the decision maker prior to the time at which the choice must be made.

Since you must decide on Oligopoly before any uncertainty resolves, as long as your choice node comes first, you are living within the strictures of that rule.^a

As for Beljeau's tree, they will learn the size of the market prior to making their decision, so the chance node for size of market must come first in their tree. And they don't learn about your decision regarding Oligopoly until after their decision, so the chance node for your decision must come after their choice node. This gives the decision tree in figure 11.3, with contributions to profit and probabilities for the market size supplied.

This is a relatively simple example, and we can see the "answer" just by inspecting the decision trees. Look at figure 11.3 first. If the market is large, then Beljeau will make a positive contribution no matter what Jokx does if they go ahead with Reaganomics, so they are pretty sure to do so. Whereas if the market is small, Beljeau is sure to lose money with Reaganomics whatever Jokx does, so they will probably decide not to proceed in this case. This means that in figure 11.1, if the market is large, you (Jokx) can be sure that Beljeau will be in the market, netting you \$10,000 if you introduce Oligopoly. Whereas if the market is small, you can be pretty sure that Beljeau will be out of the market, and you will net \$2,000 if you introduce Oligopoly. So since you are fairly sure that

^a We haven't supplied probabilities on the branches in figure 11.2, because we don't know probabilities for Beljeau's actions (yet). In general, the rule on probabilities in decision trees is that the probability that goes on any branch must be conditional on any previously resolved event, so we can't even supply probabilities for market size in 11.2(a).

Beljeau will be out if the market is small, you can safely enter the market; you will make a positive contribution no matter what.

Despite this simplicity, we will carry forward with this example. Early in the next chapter we will arrive at the conclusion just stated, and while we already know the answer to this problem, the techniques we will use will help us to see through more complex games.¹

An extensive form representation

In a *game tree*, we use one tree structure to represent *both* players' decision problems. How could we ever represent both decision problems in a single tree? The obvious problem is that the order of nodes depends on whose perspective we take. Even so, it can be done. Have a look at figure 11.4. This is rather a mess, but if you bear with me, it will become clear.

The game starts with the open circle or *node* near the middle of the page. (The game will always start at the open circle in this sort of picture.) Note that *Jokx* appears besides this open circle, meaning that at this point it is Jokx who must choose what to do. Jokx is limited to two choices that correspond to the two arrows pointing out of this initial node labeled *intro O* (for introduce Oligopoly) and *don't*.

Follow the choice *intro O* by Jokx to a second node depicted by a closed circle. (Open circles will be used to denote only the initial node.^b) This node is labeled *nature*, meaning that at this point we imagine the

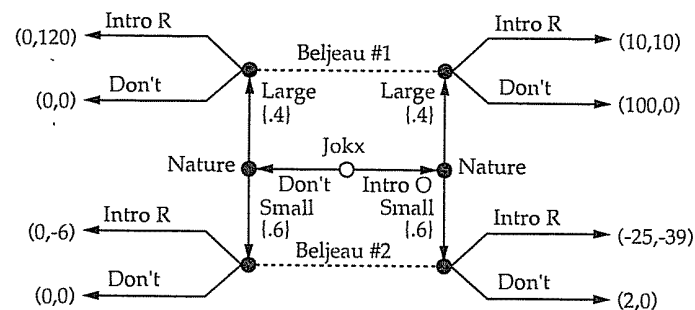


Figure 11.4. The game in extensive form.

¹ Problem 1 gives a good example. If you like to work out problems as you read along, now is a good time to do problem 1(a).

^b Or nodes — see below.

choice of which path to follow is determined by events under the control of neither player. Nature chooses between a *large* market and a *small* one with the probabilities given on the branches. You next come to nodes at which Beljeau must choose between *intro R* (introduce Reaganomics) or *don't*, and at the end of each path through the tree you find a vector of *payoffs*, in this case measured by contributions to profit. Jokx' payoff is first and Beljeau's second. So, for example, if Jokx introduces Oligopoly (move right), the market is small (move down), and Beljeau introduces Reaganomics, you reach the vector $(-25, -39)$, meaning that Jokx loses \$25,000 and Beljeau loses \$39,000. Similarly, if Jokx introduces Oligopoly, the market is large, and Beljeau doesn't introduce Reaganomics, you reach the vector $(100, 0)$, meaning that Jokx has made \$100,000 and Beljeau has netted zero.

The one "problem" is that we've put Beljeau's decision after Jokx' decision in the tree. Beljeau, remember, doesn't know what Jokx has done when it must make a decision. We depict this in the picture by the two dashed lines. Take the upper one. This connects Beljeau's two decision nodes in the circumstances (a) Jokx introduces O and the market is large, and (b) Jokx doesn't introduce O and the market is large. Beljeau cannot distinguish between these two circumstances, and so we join these with the interpretation: Whatever Beljeau does at one of the two nodes joined by the dashed line, it must do precisely the same thing at the other. This is called an *information set*; in general, all the nodes in an information set represent circumstances at which some player is called upon to (move) without knowing which of those circumstances pertain.

Recall that we labeled the initial node *Jokx*, since Jokx moved there, and the second two nodes *nature*, since the moves there were taken by forces outside of players' control. But we didn't label the next set of nodes directly with Beljeau. Instead, we labeled the two dashed lines or information sets, and these labels on dashed lines are meant to imply that at all four of the nodes in the two information sets Beljeau chooses what to do.

Most importantly, note that Beljeau has two information sets labeled *Beljeau #1* and *Beljeau #2*. These correspond to the two different decisions that Beljeau must make: Whether to introduce Reaganomics if the market is large (#1); and whether to introduce Reaganomics if the market is small (#2).

There is, of course, a strong connection between the extensive form game tree in figure 11.4 and the two decision trees in figures 11.1 and 11.3. In particular, every choice node in the decision tree of a player will correspond to one information set for the player in the game tree. So, just as there are

two decision nodes for Beljeau in 11.3, so there are two information sets in 11.4. And just as there is one decision node for Jokx in 11.1, so there is one information set for Jokx in 11.4, where a single node belonging to a player that is not joined to any other by a dashed line is thought of as an information set in its own right.

Why did we put the nodes in the order we did? In many cases, we have a lot of freedom in choosing the order. The only rule that we must follow is that

If someone learns something prior to making a particular decision, where the something learned could be either a move by nature or a move by another player, then the node representing what is learned must precede the node where the decision is taken.

Hence the node(s) for nature must precede Beljeau's nodes. Compare with the rule for decision trees, where the rule is more definitive; there we say that something outside the control of a player, depicted by a chance node, comes before a choice node *if and only if* the player learns the resolution of that chance event before making the decision. For game trees the implication runs only one way: We are allowed to put a "chance node" before a "choice node" if the player who is choosing doesn't learn what happened at the chance node until after the choice must be made; to repair this misordering, we use information sets.

In this particular example, we have three generations of nodes: One for Jokx' decision, one for Beljeau's, and one for nature's. The rule says that in a game tree nature's nodes must precede Beljeau's. But that is the only implication of the rule. Hence besides the order $J \rightarrow n \rightarrow B$ that we used in figure 11.4, it is possible to depict this situation in a game tree with the order $n \rightarrow B \rightarrow J$ and in the order $n \rightarrow J \rightarrow B$. It is probably a good exercise to draw these two "other" orders for the game tree. It hardly needs saying, but the point of this exercise is entirely in getting the information sets drawn correctly.

There is an important point here. In representing the game on a single tree, what we want to do is convey what each party knows whenever it is that party's move, and we want to include all the complete "sequences" of steps that might get us from start to finish. The word "sequence" is in quotes because there is no particular temporal sequence to things — "sets" of events might be better. Actual calendar time matters only insofar as it determines (to some, but not complete, extent) what one party might know when it is time for that party to move. That is, barring prescience,

which we will do, one can only know things that happened before. But there is no reason to suppose that one knows everything that happened before — that is hardly descriptive of reality — and it is knowledge and not time that rules what orders of moves can/cannot be used in this sort of tree.

(Once you understand this example, a useful way to test your understanding is to proceed to the problems at the end of the chapter.)

11.2. Games in extensive form: Formalities

The tree representation of a game in figure 11.4 is called a game in *extensive form*. Now that we have an example to bear in mind, we will look at the formal structure of games in extensive form. What follows is formal and mathematical, and the mathematically unsophisticated reader may find it hard to read through. Most of the examples we'll look at will be fairly straightforward, and if you understood the example in the previous section, the formalities to follow are likely to be unnecessary. But it is probably worthwhile to give what follows a try, if not to get all the details (which become fairly obscure toward the end) then at least to get a sense of how one proceeds generally. We will restrict attention in these formalities to finite player, finite action, and finite tree games, with a few comments at the end about extensions to more general settings.

An extensive form representation of a noncooperative game is composed of the following list of items: a list of players, a game tree, an assignment of decision nodes to players or to nature, lists of actions available at each decision node and a correspondence between immediate successors of each decision node and available actions, information sets, an assignment of payoffs for each player to terminal nodes, and probability assessments over the initial nodes and over the actions at any node that is assigned to nature. This is a long list and these things are subject to various rules and regulations, so we proceed to develop this list a step at a time.

(1) *There is a finite set of players.* We will number the players with arabic numerals, as $1, 2, \dots, I$, for a game with I players. A typical player will be labeled i . In addition to these players is a "natural force," which we will call *nature*, that acts something like a player. We will refer to nature with the symbol N .

(2) *There is a game tree.* This consists of a finite set T and a binary relation $\prec$ on T . A typical element of T is denoted by t and is called a *node*. The binary relation is called *precedence*; and T forms an *arborescence* with $\prec$.

This means that $\prec$ is asymmetric, transitive, and satisfies the following additional property:

If $t \prec t''$ and $t' \prec t''$ and if $t \neq t'$, then either $t \prec t'$ or $t' \prec t$.

To understand what this means, make the following constructions and definitions.

For $t \in T$ write $P(t) = \{t' \in T : t' \prec t\}$ and call $P(t)$ the set of *predecessors* of t .

For $t \in T$ write $S(t) = \{t' \in T : t \prec t'\}$ and call $S(t)$ the set of *successors* of t .

Let W denote the set of nodes in T with no predecessors, or $W := \{t \in T : P(t) = \emptyset\}$. We sometimes call W the set of *initial nodes* of the game tree.

Let Z denote the set of nodes in T with no successors, or $Z := \{t \in T : S(t) = \emptyset\}$. The nodes in Z are called the *terminal nodes* of the game tree.

Let X denote the set of nodes that are not terminal nodes, or $X := T \setminus Z$. These nodes are sometimes referred to as the *decision nodes* of the game for reasons that will become clear in a bit.

Now let us try to make sense of the requirements that $\prec$ is asymmetric and transitive and has the extra property listed above. First, we will translate this extra property into words. What it says is that if two distinct nodes precede a third then one or the other of the two nodes precedes the other one. That is to say, *the set of predecessors of a given node is totally ordered by $\prec$* , where a set is totally ordered by a binary relation if any two distinct elements of the set are ordered in one direction or the other.

Think in terms of the pictures we were drawing in the previous section, where nodes (elements of T) are depicted by circles and arrows point from some nodes to others. We say that $t \prec t'$ if, starting from t , you can find a path of arrows that point from t to some t_1 , from t_1 to t_2 , from t_2 to t_3 , and so on, until t' is reached. Note that the way this has been defined $\prec$ is automatically transitive. Asymmetry of $\prec$ then tells us that there are no cycles in this arrangement of nodes and arrows. And to say that $\prec$ totally orders the set of predecessors of a given node is to rule out the sort of thing shown in figure 11.5(a), where the node labeled x has two predecessors that are unordered by precedence.

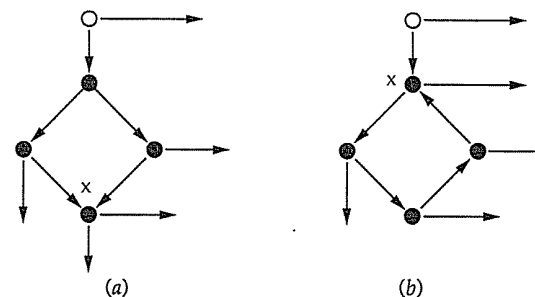


Figure 11.5. Two things that aren't allowed in game trees.

In (a) the node x has two (immediate) predecessors, neither of which precedes the other. This violates the condition that the predecessors of any node must be totally ordered by $\prec$. In (b) a cycle starts at x .

To see more formally what this assumption does, consider the following proposition:

Proposition. For each $t \in T$ which is not in W , there exists a *unique* node $p(t) \in T$ such that $p(t) \prec t$ and, if $t' \prec t$ and $t' \neq p(t)$, then $t' \prec p(t)$.

We call $p(t)$ the *immediate predecessor* of t . Uniqueness of $p(t)$, if it exists, is easy to show. If there were two nodes, say t' and t'' that satisfied this definition, then $t' \prec t''$ and $t'' \prec t'$ would both be required, and this would contradict asymmetry of $\prec$. As for existence, since $P(t)$ is a finite set and since it is totally ordered by a transitive and asymmetric $\prec$, there is some element of $P(t)$ that does not precede any other element of $P(t)$. (Prove this, if you like little logical puzzles.) This element is $p(t)$.

This is what we get from the assumption that $\prec$ totally orders the predecessors of any node t : Any node that has predecessors has a unique immediate predecessor. So if we begin at any node t , either it is an initial node or it has a unique immediate predecessor $p(t)$. And either $p(t)$ is an initial node or it has a unique immediate predecessor $p(p(t))$, and so on. Beginning from t , we can trace back along a *unique* path to some (unique) initial node. (Remember that T is a finite set, so we have to come to an initial node eventually if we are to avoid cycles.) This explains why this sort of structure is called an *arborescence* or a *tree*. Think of a point in a (real) tree. If we go back along the tree's branches from this point toward the roots of the tree, a single path of branches leads back to the trunk and

thence to the roots. Put the other way around, starting from the roots the tree opens out; it doesn't grow back in on itself.

Note that we have not insisted that our tree T have a single initial node. That is, we allow more than one node to be in W . In figure 11.4, we had a single initial node (the open circle), but we wish to keep available the possibility that the tree T has a number of possible starting points. So, to pursue the botanical metaphor, perhaps we should call T a "grove" instead of a tree. (We will stick to conventional language and use "tree.")

In fact, let us tie together carefully the sort of picture in figure 11.4 and our mathematical construction of a tree. In our pictures, nodes are of three types: open circles, filled-in circles, and vectors of numbers, with arrows pointing from some nodes to others. In these pictures, there is never more than one arrow pointing towards any given node. Open circles are distinguished by the fact that no arrow ever points toward them, i.e., arrows only point out. And vectors of numbers only have a (single) arrow pointing toward them, i.e., no arrows leave them. All nodes that are not vectors of numbers have at least one arrow pointing out from them. And no path that follows arrows is a cycle. With these geometrical conventions, we have that the open circles are the initial nodes (members of W) and vectors of numbers are the terminal nodes (members of Z). One node precedes another if some path from the first along the direction of our arrows leads to the second. The crucial properties are that each node has at most one arrow pointing at it, which corresponds to the assumption that the set of predecessors of a given node are totally ordered by precedence, and the absence of any cycles, which is asymmetry of precedence. (If you have good geometrical intuition, you can see how these properties imply that starting from any node there is a unique path back along the arrows to some node that has no arrow pointing at it.)

This says how we would create the mathematical objects of T and $\prec$ from a picture. To reverse the process, if we had a pair $(T, \prec)$, we would

(a) Draw on a piece of paper a symbol for each element t of T : an open circle if $t \in W$, a set of parentheses (to be filled in later) if $t \in Z$, and a filled-in circle for all other t .

(b) Then for each node t that has at least one predecessor draw an arrow from its unique immediate predecessor $p(t)$ to t . Done!

We will find it useful to have yet another piece of notation. For any node $t \in X$ (any node that is not a terminal node), we write $s(t)$ for the set $\{t' \in T : t = p(t')\}$. That is, $s(t)$ is the set of *immediate successors* of t .

(3) One player (or nature) is assigned to each decision node. This is given formally by a one-to-one function $\iota : X \rightarrow \{N, 1, 2, \dots, I\}$. The interpretation is that at node x player $\iota(x)$ selects what happens next. Note that we allow the possibility that nature makes this selection. If you refer to figure 11.4, you see an example: The initial node of the game (the open circle) is labeled *Jokx*, which means that Jokx moves there. Both successors to the initial node belong to nature, and so ι at each of those nodes would be N . And at each of the four successors to N 's two nodes, the move belongs to Beljeau.

(4) For each $t \in X$, there is a finite set of *available actions* $A(t)$ and a function $\alpha : s(t) \rightarrow A(t)$ that is one-to-one and onto. The idea here is that each arrow or step in the game is given a name, this being the action that is required to move from one node to one of its successors. The requirement that α is a one-to-one and onto function from $s(t)$ to $A(t)$ means that from a given node t there is a correspondence between actions that can be taken and the next node in the game that will be reached. Every action leads to a distinct successor to t , and every successor to t is reached by a distinct action.^c In terms of figure 11.4, from the node in the left-center of the page following the action *don't* by Jokx, nature has a choice of two actions, *large market* and *small*, which lead to two different nodes (where Beljeau moves).

(5) The nodes $t \in X$ are partitioned into *information sets* with the following three requirements for any pair t and t' that are in the same information set:

(a) $t \notin P(t')$ and $t' \notin P(t)$;

(b) $\iota(t) = \iota(t')$; and

(c) $A(t) = A(t')$.

The idea here is that certain subsets of decision nodes are such that the player choosing the action at one of the nodes doesn't know which of those nodes he or she is at. Refer to figure 11.4. All the decision nodes there are partitioned into five information sets. The initial node, belonging to Jokx, is by itself (as a singleton set) an information set. So are its two successors, which belong to nature. (Conventionally, nodes that belong to nature are always singleton information sets, a convention we will follow throughout.) Then there are two nodes in the information set *Beljeau #1* and two in the information set *Beljeau #2*. As shown in this figure, we join

^c If we were being very formal, we would subscript α by t , but we will not be quite that formal.

nodes in a single information set with a dashed line. (Other authors will put all the nodes of a single information set into a “balloon.”)

Information sets are meant to model the notion that a player when called upon to move might not know everything that happened “previously” in the game. (We put previously in quotation marks, because, as we noted before, the order of nodes in a game tree needn’t necessarily follow temporal order of the moves.) But built into the formal construction that we use are some restrictions to this general notion. First, we require that information sets partition the set of decision nodes. That is, every decision node is in one and only one information set. The point is that if at node t player $\iota(t)$ doesn’t know whether he is at t or at some t' (so t' is in the same information set as t), then at t' , this player doesn’t know that he isn’t at t . Now one can imagine circumstances in which a player at one point in a game is so badly confused that he doesn’t realize that he isn’t at some different point, whereas if he were at that second point, he would realize that he wasn’t at the first point. Such circumstances cannot be modeled directly in conventional game theory. (There are ways to model this sort of confusion indirectly within the structures of conventional game theory; see chapter 13 concerning how one models “irrational” behavior.)

Beyond this basic assumption, we make the three further assumptions listed previously. Assumption (a) says that a player remembers at any point in the play of a game whether he moved previously. Assumption (b) says that any two nodes in an information set have the same player moving. This means that a player doesn’t confuse a situation in which he is given the choice of action with another situation in which another player has the choice. And assumption (c) says that a player is faced with the same available actions at any two nodes from the same information set; otherwise (the logic goes) the player could look at the sets of available actions and distinguish the two nodes.

In a sense, this construction of information sets and the three assumptions reflect an underlying basic assumption of the model, namely that the structure of the game is understood by the players. Pictures such as figure 11.4 are meant to be available to each player, and each player is smart enough to look at the picture and discern from it all the implications of the structure laid out there. There is more to the three assumptions than this; for example assumption (a) includes the assumption that players remember *that* they moved previously. But it is the underlying basic assumption that plays the largest role in these formal restrictions. Now one may well complain that there are many competitive situations in which players are somewhat ignorant of pieces of the structure of the game. A player, for example, may not know some of the options available in a particular

situation. We handle this sort of thing by saying that the player in this case doesn’t really have those options. A harder case to domesticate follows: In a given situation, a player believes he does not have certain options. There is another situation that he cannot currently distinguish from the present situation, and thus he currently believes that in this other situation he would not have those options. But in this second situation he would, in fact, know that those options are available, and so he can distinguish between the two. This we can’t model directly. (It is hard enough to say.^d) Or to take a third example, there may be circumstances in which your rival has available options that you don’t know she has. This we can’t model directly. We will discuss in chapter 13 how we might try to model these things; for now understand that they are things seemingly outside the bounds of our formal models.

We will use H to denote the set of all information sets in a given game, with h a typical information set. That is, H is a partition of X and each h is a subset of X . By virtue of (b) and (c), every node t from a given h has the same assignment of player and the same set of available actions. That is, if $t, t' \in h$, then $\iota(t) = \iota(t')$ and $A(t) = A(t')$. Accordingly, we will abuse notation and write $\iota(h)$ for the player whose turn it is to move at all nodes in h and $A(h)$ for the set of actions available to $\iota(h)$ at any node in h . Finally, for any $x \in X$, we will write $h(x)$ for the information set that contains x .

In addition to the three assumptions above, it is sometimes conventionally required that for t and t' that are in different information sets, $A(t) \cap A(t') = \emptyset$. This assumption is useful to keep notation from getting out of hand and it is without any loss of generality, so we make it whenever convenient.

(6) An assignment of *payoffs* for each player to each terminal node of the game is given. Formally, a function $u : \{1, 2, \dots, I\} \times Z \rightarrow R$ is given, where $u(i, z)$ is the payoff to player i if the game ends at node z . We sometimes will write this function with the i as a subscript as in $u_i(z)$. In our pictures, we write vectors at each terminal node for the vector of payoffs, with player 1’s payoff the first component of the vector, and so on. Insofar as there is or will be any uncertainty in the game, we think of the players as maximizing expected utility, and we think of the payoffs as denominated in units of utility. Insofar as the players pick up bits of payoff as the game

^d Develop a formal model of this situation. The key is to have separate formal objects for the “objective reality” and for a player’s subjective perceptions of reality. Information sets formalize subjective perception, but the conditions placed on information sets limit what can be modeled with them.

goes along, all those bits are aggregated into a final outcome and then translated into units of utility by the function u .

(7) A probability distribution ρ is given over the set of initial nodes W , and for each node $t \in X$ with $u(t) = N$ a probability distribution ρ_t over $A(t)$ is given. We think of nature as determining which initial node of the game pertains and then, perhaps, moving during the course of the game. Nature's choice of initial node is made according to the probability distribution ρ . If called upon to move during the course of the game, say at node t , the probability that nature takes action $a \in A(t)$ is given by $\rho_t(a)$. In our pictures, we will represent these probabilities by numbers in curly brackets, either at the initial nodes of the game, as in figure 11.7 (a few pages further on), or along the branches leading out of nodes that belong to nature, as in figure 11.4. (When there is a single initial node for the game, as in figure 11.4., we will neglect to put the obvious probability 1 in a curly bracket.)

We make two points about these probabilities. First, it is conventionally assumed that all players share the same assessments over nature's actions. This convention follows from deeply held "religious" beliefs of many game theorists. (Recall our discussion of the Harsanyi doctrine in chapter 3.) Of course one hesitates to criticize another individual's religion, but to my own mind this convention has little basis in philosophy or logic. Accordingly, one might prefer, being more general, to have probability distributions ρ and ρ_t , which are indexed by i , reflecting the possibly different subjective beliefs of each player. For most of this book, we won't consider examples for which this additional structure will be required, and so we follow the conventional assumption except when noted. But in following convention on this point, I do not mean to endorse the dogma that lies behind it.

Second, most treatments of this subject either look at games where W is a singleton set or at games where nature is never assigned to any node. The idea is that if W has more than one element, we could draw in a "super-initial node" that precedes the nodes of W , give nature the move at this super-initial node, and have her pick one of the initial nodes from the original W . Alternatively, in any game where nature is given the move at some node, we could remove that node by having more initial nodes. Both ways of doing things are without loss of generality; with either one (alone), you don't need the other. But we will allow here for both, since sometimes the pictures are neater one way, and sometimes they are neater the other.

That is it. (That's not enough?) That is how, formally, one describes an extensive form game.² As noted above, this sort of formalism does put some restrictions on what you can capture, but a lot of things still can be represented. Five things deserve special mention here.

Simultaneous moves

Imagine a game with two players, 1 and 2, who must make their (decisions) simultaneously. Specifically, player 1 must choose between L and R, whereas player 2 must choose between l and r. Then payoffs are made; for example, if player 1 chooses L and 2 chooses r, player 1 receives 2 and player 2 receives 3. These decisions, because they must be made simultaneously, are each made in ignorance of what the other player is choosing.

We can represent this in an extensive form game as in figure 11.6(a) or as in figure 11.6(b). Note that in (a) we have the node for player 1 coming before the two nodes of player 2, while in (b) we have the node for player 2 coming before the two nodes of player 1. But in each case, we use an information set to record the fact that the player "moving second" moves without knowing what the "first mover" did. Note also the positioning of the payoff vector (2,3) in these two figures; in (a) it comes after the sequence L-r, while in (b) it comes after r-L.

At the risk of beating this to death, note that figures 11.6(a) and (b) are both also consistent with the story. Player 1 chooses between L and R at 9:00 A.M., and player 2 chooses between l and r at 9:10 A.M., but player 2,

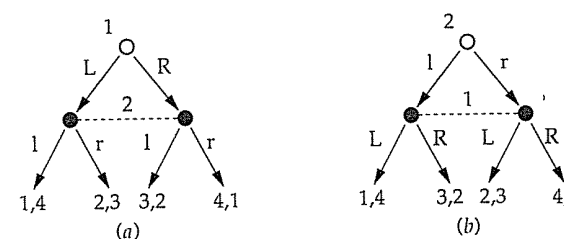


Figure 11.6. Two extensive form representations of a simultaneous move game.

² A piece of terminology you might run into is that all the pieces except the last two (except for payoffs and probability assessments for nature) taken together are referred to as an extensive form game form.

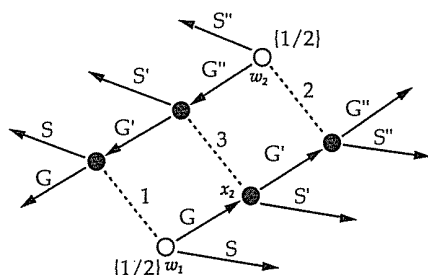


Figure 11.7. An extensive form game with a strange order of moves.

when he chooses, doesn't know what player 1 chose. Insofar as no choices are simultaneous, it wouldn't hurt to use physical time to determine precedence, but precedence in extensive form games isn't meant necessarily to convey the temporal sequence of events. As such, there is no problem in representing even simultaneous choices of action.

Precedence and information sets

Consider the following situation and the game tree in figure 11.7. The three participants are called 1, 2, and 3. Either 1 or 2 is asked to choose first, between giving the move to 3 (by choosing G or G') or ending the game (by choosing S or S'). If 3 is given the move, she can either end the game (by choosing S') or give the move to 2 or to 1 (by choosing G'). Then, if they get the move, 2 or 1 chooses between two actions (G or S for 1, and G' or S'' for 2). The chance that 1 is asked to move first is 1/2. (Note that we are using multiple initial nodes in this picture.)

The tricky part of this example is that each of the two initial nodes belongs to an information set for the player who moves there that also includes the other node that belongs to that player. In other words, player 1, if called upon to move, doesn't know whether she is moving first, because nature started the game with her on the move, or third, because nature started the game with 2 on the move, who gave the move to 3, who gave it to 1 — and similarly for 2. And 3, if given the move, doesn't know whether 1 moved first or 2 did.

In the game, there is a well-defined order of moves at the level of nodes: A clear succession of nodes (and actions) leads to each terminal node. But at the level of information sets, which is the level of what the players are assumed to know, the order of moves can be uncertain. In

particular, we can say that the initial node labeled w_1 in the tree precedes the node labeled x_2 . But the information set to which w_1 belongs, namely 1's information set, cannot be said either to precede or to follow 2's or 3's information set.

For what we will do in this book, there is no need to insist on a precedence order for information sets. But for some theoretical developments such an ordering of information sets is useful. For this reason, there is a specialization of the general notion of an extensive form, due to von Neumann, in which information sets can be ordered — if you ever encounter the term “an extensive form in the sense of von Neumann,” it is (probably) to this that the author refers.³

Perfect recall

Look next at the tree structures in figure 11.8. Each is peculiar. Figure 11.8(a) is peculiar in that player 1 moves first and then, if she gets to move again, her information set indicates that she forgot what she did. That is, she is assumed to be unable to distinguish between the three sequences of actions (l, R) , (r, L) , and (r, R) . We have no problem with her being unable to distinguish between the second two of these sequences; it is player 2 who chooses between R and L , and it is quite reasonable to suppose that player 2's choice is not revealed to player 1. But how can player 1 fail to distinguish between (l, R) and those other two? If she remembers that she chose l instead of r , she won't be confused.

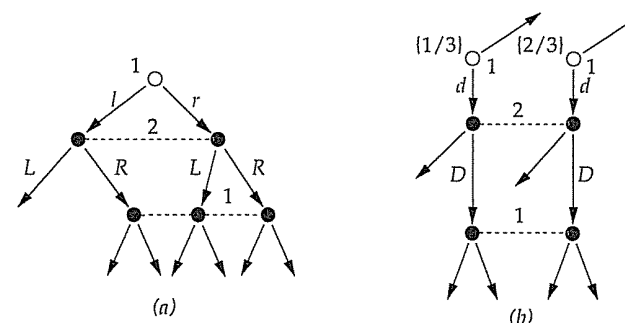


Figure 11.8. Two types of “forgetful” player.

³ To be precise concerning the development of these ideas, von Neumann's extensive form came first and was later generalized to games in which information sets may not be ordered by precedence.

Figure 11.8(b) is interesting in that player 1 is modeled as making her first choice aware of which node nature chose to begin the game. This is indicated by the fact that the two initial nodes are not joined by a dashed line; each as a singleton set is an information set in its own right. But if 1 chooses d (from either starting point) and then 2 chooses D , 1 moves again, and the dashed line indicates that 1 seems to have forgotten what nature did, something she knew previously.

These two examples illustrate that none of the assumptions made so far preclude a player from forgetting certain things as the game progresses. A player is not allowed to forget the fact that she moved previously; assumption (a) concerning information sets precludes this. But a player is permitted, in general, to forget what she did earlier, and she may forget things she knew when moving earlier.

In the analysis of many games, it is useful to assume that the players don't forget these two sorts of things. Accordingly, we say that a game has or satisfies *perfect recall* if players remember what they previously knew and what they previously did.

Here is how this assumption is written formally.

Definition. Given a game with information sets H , the game is said to satisfy or have *perfect recall* if, for any three nodes x , x' , and x'' with $i(x) = i(x') = i(x'') = i$ such that $x \prec x'$ and $h(x') = h(x'')$, then

(a) $P(x'') \cap h(x)$ is a singleton set. We use x^0 to denote the node in this intersection. (The case where $x^0 = x$ is definitely not precluded.)

(b) Let $\{y^0\} = s(x^0) \cap [P(x'') \cup \{x''\}]$ and let $\{y\} = s(x) \cap [P(x') \cup \{x'\}]$. Then $\alpha(y^0) = \alpha(y)$.

Part (a) is, roughly, that players remember everything they knew before. Since x' and x'' are in the same information set, and since x precedes x' , there must be some node preceding x'' at which point the player's information was just the same as her information at x . And, from those two points, since she remembers what she did, she must have taken the same action (which is part [b]).

If this definition is mysterious, the reader who is interested in formalism should know that this definition is simpler than the one originally given. The classic reference on games with perfect recall and indeed for the construction of extensive form games is Kuhn (1953). Kuhn does things a bit differently than is done here: In particular, his definition of perfect recall is a bit weaker and even less transparent than our definition. Since Kuhn's definition is what is needed for his theorem (cf. section 11.4), it is certainly the "right" definition mathematically. But because it is a bit harder to understand, we use this stronger rendition. The restriction on information sets — if $x \prec x'$, then $x \notin h(x')$ — is sometimes put as part of the definition of perfect recall instead of being imposed as part of the definition of an extensive form game. It

is included here as part of the definition of a game because I know of no useful or interesting examples where it is violated, whereas there are some interesting examples of games that satisfy it but that violate perfect recall as defined here; cf. problem 5. It should be noted, finally, that perfect recall as defined here implies this first restriction on the structure of information sets.

We noted before that, in general, information sets cannot be ordered by precedence. If a game has perfect recall and if we look at two information sets for a single player, however, either every node in one has a predecessor in the other (and not the reverse), or no node in one has a predecessor in the other. In fact, for any game with perfect recall (or even for a game in which some given player i has perfect recall), one can construct from the game tree a *decision tree* for the player where each choice node by the player corresponds to one of the player's information sets and where chance nodes correspond to combinations of actions of other players. The serious reader may find it helpful to give the details of this construction.

Infinitely many players and/or actions

When there are infinitely many players in the game or infinitely many actions (including the possibility of infinitely many initial nodes or moves by nature), then the formal objects given above will not suffice. For example, in some games infinitely many players move simultaneously. Having one move "first," a second move second without knowing what the first did, and so on, will not present a very pleasant looking tree. So, to handle infinities the mathematical notion of an extensive form game must be "tuned up." There is no best or most general way to do the tuning; for special cases, there are specially designed types of extensive games. Aumann (1964) should be consulted by the extremely serious reader.

Rigid and well-specified rules

We noted already that there are certain sorts of confusion that we cannot model with the notion of an extensive form game as described. There are other situations that would be, at the least, hard to model in the fashion given above.

Imagine a situation where two firms are linked by a contractual arrangement; think of the firms as a coal mine and an electric power-generating firm. The contract specifies, for example, that the electric power-generating firm may demand each month not more than X tons of coal at a set price p per ton but the generating firm must pay for at least Y tons of coal for some $Y < X$ even if it doesn't take the coal. Now imagine that demand for electricity falls, so much so that the electric power-generating firm knows that it will not need Y tons of coal. It is willing to pay a price

higher than p if it can be released from its obligation to buy at least Y tons. So a meeting is arranged between the two parties to see if they can come to a mutually agreeable renegotiated contract. We are interested in modeling the negotiation process.

Who moves first? Who moves second? What officially signifies that a mutually agreeable deal has been made? In most negotiations these things, which constitute the protocol of negotiations, are not really specified with the precision we require for an extensive form game.

Two options are open to us at this point. One is to forego trying to model the negotiation as an extensive form game and instead use some principle concerning how we think negotiations will conclude. An example of such a principle is: The two will come to an efficient arrangement (one in which it is impossible for both sides to be made better off) that "splits the difference" above the levels of their profits if they have to keep to the original contract. This approach would move us from noncooperative to cooperative game theory, where one looks for general principles on which "deals" will be struck given a description of what players can achieve if they cooperate and if they don't.

Alternatively, we can stick to the "rules" of noncooperative game theory, which are that we have to be very precise about the rules of the game. We might look at the negotiations as a process in which the two make alternating offers with the honor of making the first offer determined by a coin flip. We might look at the negotiations as a process in which the two must simultaneously make offers until they make offers that are compatible (meanwhile living under the strictures of the original contract). In fact, we will study *bargaining protocols* of these sorts in chapter 15. The point for now is that the methodology we are developing requires a very precise and rigidly specified protocol, which in some instances may be quite artificial.⁴

11.3. Games in normal or strategic form

From the example of section 11.1 it should be obvious that some game situations can be represented by a number of different extensive forms. Since all these extensive forms represent the same "game," we might suspect there is another way to represent the game that is a bit clearer about the essence of the situation.

⁴ We might hope for results that say the precise protocol doesn't matter. If your hopes run in that direction, chapter 15 will be somewhat disillusioning.

What is common to all the different extensive form representations hinted at for Jokx versus Beljeau is that they all represent the same strategic problem for the two sides. Suppose that as owner of Jokx you have decided to take a vacation in Florida. All decisions concerning Oligopoly must be made before you return, and you refuse to ruin your vacation by talking on the phone with the home office. So you decide to leave complete and unambiguous instructions as to what decisions you want taken regarding Oligopoly.

That's pretty easy — you have to decide whether to proceed with Oligopoly or not. You will receive no useful information, so there are really only two possible sets of instructions you might leave:

- s_1 : Proceed to market Oligopoly.
- s_2 : Don't do this.

What is more interesting is the similar problem for the management of Beljeau. If they are headed off for vacation, the instructions they must leave will concern whether to market Reaganomics or not, contingent on the results of the market survey. There are four sets of instructions they could leave:

- t_1 : No matter what the survey says, proceed with Reaganomics.
- t_2 : Proceed with Reaganomics if the market will be large, but not if the market will be small.
- t_3 : Proceed with Reaganomics if the market will be small, but not if the market will be large.
- t_4 : No matter what the survey says, don't proceed with Reaganomics.

Now you may find strategy t_3 to be fairly silly. (In fact, at the end of section 11.1 we already indicated that t_2 seems the only sensible thing to do.) We'll worry about choosing among these four *strategies* in the next chapter. For the time being we leave them all, since they represent all the instructions Beljeau could conceivably leave behind before going off on vacation.

Look at the extensive form depicted in figure 11.9. The game depicted there runs as follows. You (Jokx) are heading for Florida, so you select one of your two strategies. Without knowing what you chose (note the information set), Beljeau picks one of their four strategies. And then nature acts to select one of the two market sizes, with payoffs made to each side accordingly. For example, if you pick s_2 , Beljeau picks t_2 , and the market is large, then you are out of the market and Beljeau learns that the market

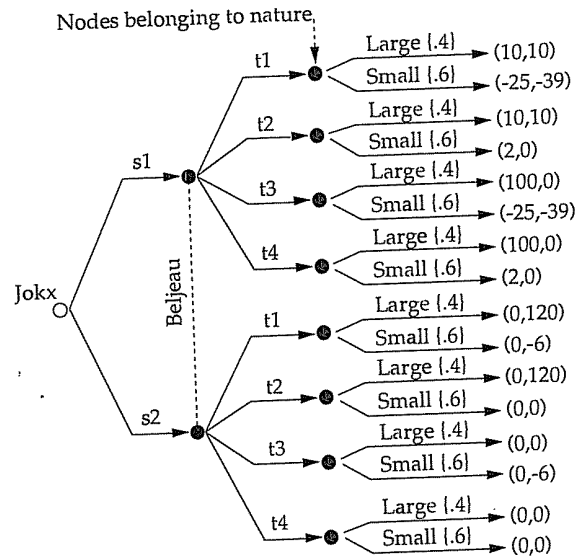


Figure 11.9. Another extensive form representation of Jokx vs. Beljeau.

is large and (according to $t2$) markets Reaganomics. Thus you make \$0 and Beljeau makes \$120K. Note that we could just as well put Beljeau first and you second, using an information set for you to model the notion that you don't know what strategy Beljeau has selected when you select your own strategy.

One final step, and we have what is known as the *normal or strategic form* of the game. Let's assume that both you and Beljeau are risk neutral — that you evaluate risky prospects according to their expected value. (In general, since endpoints in games are evaluated in terms of the players' von Neumann-Morgenstern utility functions, expected payoffs become expected utilities.) Then if, say, you pick strategy $s1$ and Beljeau picks $t2$, you stand a .4 probability of getting a contribution of \$10,000 and a .6 probability of a contribution of \$2,000. This has an (expected) value of \$5,200. Similarly, for this pair of strategies, Beljeau has an (expected) value of \$4,000.

In figure 11.10 we give what is known as the *normal or strategic form* of the game. In this case, because there are two players, this is also sometimes called a *bimatrix* form game. Each row in the table corresponds to one of

your two strategies. Each column corresponds to one of Beljeau's four. And in the cells of the matrix, we list your expected contribution and then Beljeau's for the strategy pair corresponding to the cell.

In general, a *normal or strategic form game* is given by: a list of players $i = 1, \dots, I$; for $i = 1, 2, \dots, I$, a list of strategies S_i that player i might employ; and for each I -tuple of strategies $(s_1, \dots, s_I)$, one for each player, the payoff to each player of that combination of strategies, given by functions $u_i : \prod_{j=1}^I S_j \rightarrow R$. Just the list of players and the lists of their strategies is sometimes called a *normal form game form*.

In moving from an extensive form to the resulting normal form, we undertake the following two-step procedure. First, for each player $i = 1, 2, \dots, I$, the set of strategies for player i is given by

$$S_i = \prod_{\{h \in H: u(h)=i\}} A(h).$$

That is, a strategy s_i for player i specifies precisely which action the player will take in every information set assigned to that player.

Second, (for every combination of strategies for the various players, which we will call hereafter a *strategy profile*, one evaluates the (expected) utility for each player, where one is taking expectation over any randomness in the initial node or in subsequent moves by nature, using the probability distributions ρ and ρ_t that are given as part of the extensive form game.

Consider the extensive form game depicted in figure 11.11. Note that player 1 has two information sets and player 2 has one. At player 1's first information set, she has two moves; she has two as well at her second. According to the formula just given for the strategy space of a player, she should have four strategies, viz., Aa, Ad, Da, and Dd. But if at her first information set she chooses D, there is no reason to consider what she would do at the second; by her own first action the second is precluded. In a sense, Da and Dd are the same strategies for her.

		Beljeau's strategy			
		t1	t2	t3	t4
Jokx' strategy	s1	-11,-19.4	5.2,4	35,-23.4	41.2,0
	s2	0,44.4	0,48	0,-3.6	0,0

Figure 11.10. The normal form representation of Jokx vs. Beljeau.

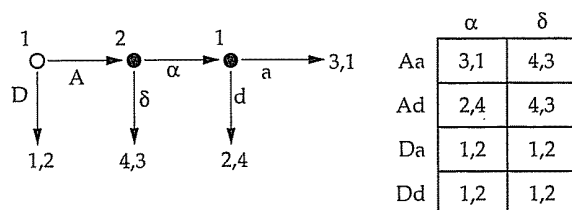


Figure 11.11. An extensive form game and its normal form counterpart.

We can see this as well if we look at the normal form of this game, also shown in figure 11.11. We have drawn the normal form with four strategies for player 1. And you will quickly note that the last two rows, corresponding to the strategies Da and Dd, are identical. No matter which strategy player 2 uses, these two strategies for player 1 give identical payoffs to each player.

In general, when two strategies by a single player give every player the same payoff for any combination of strategies by the others, we say that the two strategies are *equivalent*, and we sometimes list "them" only once in the normal form. This applies generally, but it arises most frequently in the sort of case shown in figure 11.11 where earlier actions of a player can render inconsequential her plans at later information sets, because her earlier actions mean that the later information set will not be reached. To be very precise, when we drop equivalent strategies in this fashion, we are creating what is called the *reduced normal form*.

On the other hand, there is no single way to proceed in general from a normal form game to a corresponding extensive form game. In one obvious extensive form the players all choose complete strategies simultaneously, but often other extensive forms could be constructed from a given normal form.^e

11.4. Mixed strategies and Kuhn's theorem

For reasons that will remain obscure until the next chapter, we consider in this section the possibility that a player might wish to choose her strategy by some random process.

For games in normal form the construction is easy in principle. Take a normal form game with finitely many players, each of whom has available finitely many strategies. Such a game is specified by a list of players

^e In fact, some of the early work in the theory of extensive games took on the question: For a given extensive form game, what other extensive form games are strategically equivalent, in the sense that they give the same (reduced) normal form game? For those of you who are interested, the classic papers to consult are Dalkey (1953) and Thompson (1952).

$\{1, 2, \dots, I\}$, (finite) sets of strategies S_i for each player, and payoff functions $u_i : \prod_{j=1}^I S_j \rightarrow R$. Let Σ_i be the set of probability distributions over S_i . An element $\sigma_i \in \Sigma_i$ is called a *mixed strategy* for player i , with the interpretation that if i chooses to play σ_i , then i chooses to take strategy $s_i \in S_i$ with probability $\sigma_i(s_i)$. We write s_i for the "mixed" strategy (element of Σ_i) according to which s_i is done with probability one — these are then called *pure strategies* for player i . It is assumed that players, if they mix, mix independently of one another, so that if players choose the mixed strategy profile $\sigma = (\sigma_1, \dots, \sigma_I)$, the probability that the pure strategy profile $s = (s_1, \dots, s_I)$ actually happens is $\prod_{i=1}^I \sigma_i(s_i)$. Thus if players use the mixed strategy profile σ , the expected payoff to player i is

$$u_i(\sigma) = \sum_{s=(s_1, \dots, s_I) \in \prod_{j=1}^I S_j} \left[\prod_{j=1}^I \sigma_j(s_j) \right] u_i(s).$$

While this is all very simple in principle (and we will have examples in a moment and also in chapter 12), the constructions just given do rely on the fact that there are finitely many players, each of whom has a finite number of (pure) strategies s_i . One must be careful with this sort of definition when there are either uncountably many players or players with uncountably many strategies or both. Consult Aumann (1964) if you wish to see examples of how this is done.

The subject of mixed strategies becomes more interesting when we turn to games in extensive form. Consider the game depicted in figure 11.12(a). In this game, player 1 has two pure strategies and player 2 has

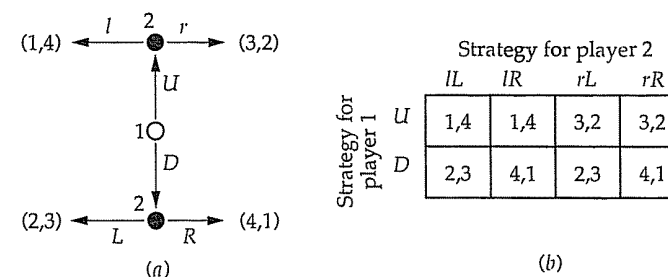


Figure 11.12. A game in extensive and normal form.